

Projective Dynamic Logo (PDL)

Global Mapping of Structures, Results, and Open Problems

Version 30 — A Guide for Continuation of the Programme

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Abstract

This document provides a systematic and self-contained guide to the Projective Dynamic Logo (PDL) programme, intended for a scientific colleague who wishes to understand, verify, or continue the research. PDL is a foundational physics programme that derives fundamental physical constants and dynamical laws from four axioms on finite signed graphs, without presupposing spacetime, particles, or fields. The minimal admissible closure under the axioms is the complete signed graph K_4 on four vertices and six edges, identified with the electron prototype. The proton is characterised by a unique integer quintuplet (24, 28, 930, 10087, 11017), from which all results flow.

The current corpus (D01–D65, plus auxiliary documents, the PDL-V series DL01–DL02, the joint PDL–OFN bridge note N01, and the note N02 in preparation) has reached the following state:

- Three critical Gates have been resolved: the axiomatic derivation of the amplitude $155/11017$ (Gate 1, D29), the structural proof that the QCD correction coefficient equals 2 (Gate 2, D30), and the theorem $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ (Gate 3, D31/D36).
- The quantum dynamics layer is **complete**: the Schrödinger equation (D32), Dirac equation (D33), Born’s rule Level 1 via Gleason uniqueness (D34), the Einstein coupling equation (D35), Born’s rule Level 2 including $U(1)$ phase freedom (D46), and the London equation (D49) have all been derived from C1–C4 with no free parameter.
- Newton’s gravitational constant G is a theorem, modulo a single irreducible external parameter ($\Delta m_{\text{iso}} = m_d - m_u$).
- The Hubble tension is resolved at 0.006% with zero free parameters (D35/D27).
- Black hole thermodynamics is **complete**: the area law $S \propto R_{\text{surf}}$ (D37), the Bekenstein–Hawking formula $S_{\text{BH}}/k_B = 4\pi(M_{\text{eff}}/M_{\text{Pl}})^2$ (D38), the one-quarter coefficient $\frac{1}{4}$ in $S_{\text{BH}} = k_B A/(4\ell_P^2)$ (D50), and the London equation (D49) are all unconditional theorems.
- Open Problem OP1 has been fully resolved by D42: the Indifference Lemma H3 is now an unconditional theorem of axioms C1–C4.
- D46 resolves OP4: the $U(1)$ phase freedom of quantum mechanics is derived from the K_4 pulsation structure, completing the quantum dynamics layer.
- D47 resolves OP13 and OP14: the spin–orbit splitting $\Delta n = 4$ is forced by a unique perfect-square discriminant (149^2), the sub-shell filling rates $r_{\text{exc}}(Z)$ are derived analytically for all $Z = 1\text{--}82$, the Mirror Lemma establishes proton–neutron HO-PDL level isomorphism, and the periodic table (valley of stability) is a corollary of C1–C4. The nuclear stability layer is **complete**.
- D48v3 fully resolves OP2-D35 and OP-spin: the coherence stress-energy tensor C_{coh} is derived with four unconditional components. $C^{(\text{mass})}$ and $C^{(\text{orb})}$ are unconditional theorems; $C^{(\text{spin})} = 0$ in the homogeneous ground state (theorem from explicit computation over all 8 coherent configurations of K_4 , resolving OP-spin); $C^{(\text{leak})}$ is a theorem via D51+D52 in dimensionally consistent units J m^{-3} . Four dimensional errors from v1/v2 are corrected.

- D49 resolves OP-London: the London equation $j_s^i = -(n_{\text{coh}} e^2 / m_p c) A^i$ is derived from C1–C4 via the metric identity $\nu = \frac{1}{4} \|\psi_1 - T^k \psi_1\|^2$ (768/768 exhaustive verification) and axiom C4.
- D50 resolves OP12/BH-3: the coefficient $\frac{1}{4}$ in the Bekenstein–Hawking formula is derived from C1–C4 as the stable fraction 4/16 per surface relation, under H3 (D42).
- D51 and D52 together resolve OP1-D35: the cosmological leakage constant is $C = (1 - \kappa)^{997} \times (930/11017)^{23} \times (1 - \eta_L)^{67}$, agreeing with the observational bound to 0.17 ppm. The causal chain $\text{C1–C4} \rightarrow K_4 \rightarrow \beta_1 = 3 \rightarrow (p_{k_1}, p_{k_2}, p_{k_3}) \rightarrow C \rightarrow \Lambda_{\text{PDL}}$ is complete.
- D53 provides a self-contained synthesis of the complete causal chain $\text{C1–C4} \rightarrow \Lambda_{\text{PDL}}$, with an independent numerical verification of C_{PDL} to 0.41 ppm relative to D51, and a systematic robustness analysis of the prime exponent structure. The chain spans fifty-seven orders of magnitude from combinatorial axioms to the cosmological constant.
- D40 derives the complete architecture of nuclear stability from the proton and neutron quintuplets alone: $N_{\min}(Z) = Z$ for $Z \leq 20$ as an exact theorem, the conflict saturation $C(Z > 20) = 190 T_{pp}$ exactly, and the valley of stability reproduced at 100% accuracy for all $Z = 1$ to 82.
- D41 confronts the PDL framework with the experimental results of Ha et al. (*Nature Communications* 16, 10631, 2025), confirming the structural identity $T \approx (\Delta n + 1)^2 = 25$ across three independent observables, with the wave-function component ratio agreeing at the 2% level.
- D43 establishes the computational infrastructure of the programme and proves the geometric leakage parameter $\varepsilon_{\text{geom}}(p) = 329/10087$ as an unconditional theorem (OP-A resolved).
- D44 derives the hierarchical filter factor $k = 0.921716$ from C1–C4 as an unconditional theorem (OP-B resolved). The complete causal chain $\text{C1–C4} \rightarrow K_4 \rightarrow \text{quintuplet} \rightarrow \varepsilon_{\text{geom}} \rightarrow \varepsilon_G = \varepsilon_{\text{geom}} \times k^{18} \rightarrow G$ is now fully closed with no remaining open problem and no free parameter beyond Δm_{iso} .
- D45 derives the first falsifiable astrophysical confrontation: the primordial black hole survival threshold is shifted upward by +11.89% relative to the standard GR value, giving $M_{\text{PDL}}^* = \sigma(40)^{-2/3} M_{\text{GR}}^* \approx 5.706 \times 10^{14} \text{ g}$, testable with existing Fermi-LAT data via BLACKHAWK and ISATIS.
- **New in v22 — D54** resolves OP-pressure [1]: the equation of state of the PDL coherence fluid is $w(N) = -\rho_\Lambda / \rho_{\text{coh}}(N)$, an unconditional theorem of C1–C4. In the homogeneous ground state, $w_{\text{mass}} = w_{\text{spin}} = w_{\text{orb}} = 0$ exactly (three distinct theorems), and $w_{\text{leak}} = -1$ intrinsically. The PDL dark-energy fraction $\Omega_\Lambda^{\text{PDL}} = \rho_\Lambda / \rho_{\text{crit}} = 0.6838$ agrees with the Planck 2020 value $\Omega_\Lambda^{\text{obs}} = 0.685$ to 0.17%, with no cosmological parameter as input. The PDL coherence fluid interpolates between pressureless cold matter at sub-nuclear scales and dark energy at cosmological scales — both emerging from the same combinatorial structure with no parameter adjustment.
- **New in v23 — D55** resolves OP10 at the level of the Weinberg angle [2]: the formula $\theta_W = 19\pi/119$ is derived as an unconditional theorem of C1–C4, via four established results (D33, D46, D51, and the new Lemma D) and zero free parameters. The key structural result is that θ_W is a dephasing sequencing ratio: the Berry phase accumulated during $k_2 = 19$ elementary interface steps out of $N_{\text{total}} = 119$, where $k_2 = n_u - R_e + 1$ (D51 theorem) and $N_{\text{total}} = (n_d/4) \cdot (r_d - r_u)/R_e = 7 \times 17$ (Lemma D, proved exhaustively over all 768 $(A) \wedge (B)$ configurations). The predicted $\sin^2 \theta_W = 0.231196$ agrees with the PDG 2024 value 0.23121 ± 0.00003 at 0.48σ . As a secondary result, combining with the geometric zeroth-order formula yields a new PDL prediction $\Delta m_{\text{iso}} = 2.446 \text{ MeV}$ (within 0.92σ of FLAG 2024), falsifiable by next-generation lattice QCD. The electroweak sector of the Standard Model is now structurally derived from C1–C4.
- **New in v24 — D56** resolves OP-D41-1-A [3]: the number of maximal partial closures on a $k p$ - $k h$ nuclear interface surface is $N_{\text{comp}}(k) = k$ exactly, proved from three lemmas (uniqueness on one unit, disjointness of mixed-triangle sets, C3-irreducibility of the assembly) and verified for $k = 1, \dots, 5$.
- **New in v28 — D61** resolves OP-D59-2 [4]: the minimal covariant derivative $D_\mu = \partial_\mu - ig A_\mu$ is derived from C4 as a survival theorem. The group of C4-admissible discrete transports on K_4 is $\text{Coh}(K_4) \cong \mathbb{Z}_2^3$ (2048 exhaustive checks); V_4 acts faithfully on this group by automorphisms

with fixed-point subgroup $F \cong V_4$ labelled by the three perfect matchings of K_4 (288 checks); the equivariant connected completion is $G_{\text{gauge}} = \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$, consistent with D46–D60; and Utiyama’s uniqueness theorem identifies D_μ as the unique first-order equivariant generator. The gauge sector of the Standard Model — group, representations, and gauge dynamics — is now completely derived from C1–C4, without variational postulate and without free parameter at the structural level.

- **New in v28 — D60** [5] proves Hypothesis $H_{\text{SU}2}$ as an unconditional theorem of C1 and C2 alone (OP-D57-1 resolved). The V_4 -invariance of the pulsation structure forces $G_{\text{eff}} = S_4/V_4 \cong S_3$, establishing the complete logical chain from the axiom of existence (C1) to the effective gauge symmetry group, with no conjectural step remaining in the gauge sector.
- **New in v25 — D57** [6] provides the first combinatorial derivation of the tree-level Weinberg angle from C1–C4 alone: $\sin^2 \theta_W^{(\text{tree})} = \sin^2(\pi/R_e) = 1/4$, where $R_e = 6$ is the relational budget of K_4 (number of edges), itself a theorem of C1–C4 (D16a). Two independent routes are proved equivalent: the combinatorial route ($192/768 = 1/4$ from the $(A) \wedge (B)$ criterion, D29) and the orbit route ($\dim(\mathbf{1})/\dim(\text{orbit-4}) = 1/4$ from the S_4 action on coherent configurations). Both routes identify the factor $1/4$ with $|V_4|/16$. The quotient $A_4/V_4 \cong \mathbb{Z}_3$ is identified as the combinatorial origin of $\beta_1(K_4) = 3$, providing a partial resolution of OP-OFN-1 (N01 [7]).
- **New in v25 — N01** (Laubscher & Evdokimov) establishes the PDL–OFN bridge [7]: $\beta_1(K_4) = 3$ and $b_1(\Omega_{21}) = 3$ under Hamming distance-1 adjacency are the same topological invariant viewed from two independent frameworks. Five Python scripts verified the convergence exhaustively. Open Problem OP-OFN-1 (formal derivation of the link between the three PDL leakage cycles and the three OFN fermion generations) is identified; partial resolution provided by $A_4/V_4 \cong \mathbb{Z}_3$ (D57).
- **New in v22 — PDL-V programme (DL series)**: Documents DL01 [8] and DL02 [9] inaugurate a structurally distinct extension of the PDL framework to the threshold conditions for life and consciousness. DL01 introduces the self-replication operator S and establishes that $S(K_4) \cong K_4$ (8/8 coherent configurations), that $\delta^* = 1/2$ is an invariant, and that the rarity law $f(n) = 2^{n+1-\binom{n}{2}}$ holds exactly. DL02 proves three analytical theorems: K_4 is the unique fixed point of S (Theorem 1); the life threshold strictly precedes the consciousness threshold, $n_{\text{life}}^* < n_{\text{consciousness}}^*$ (Theorem 2); and both thresholds are finite, $n_{\text{max}}^* < \infty$ (Theorem 3), giving the hierarchy $4 < n_{\text{life}}^* < n_{\text{consciousness}}^* < n_{\text{max}}^* < \infty$ as an unconditional theorem of C1–C4. These documents are presented in Part III of this mapping, clearly separated from the physical constants programme of Part I and II.
- **New in v27 — D59** constructs the fundamental representation $\mathbf{3}$ of $\text{SU}(3)$ from C1–C4 and identifies the physical colour gauge group $\text{SU}(3)_c$. The carrier space is the trace-zero plane $W = \{(a, b, c) \in \mathbb{C}^3 : a + b + c = 0\}$, whose axes are canonically labelled by $V_4 \setminus \{e\}$. Five lemmas verified by exhaustive exact-integer computation (PDL_D59_script1.py [10]): (L1) S_3 -action on $V_4 \setminus \{e\}$ faithful and transitive; (L2) $\chi_\pi = \chi_{\text{trivial}} + \chi_{\text{standard}}$; (L3) $W = \{a + b + c = 0\}$ unique S_3 -invariant complement of $\mathbb{C}(1, 1, 1)$; (L4) A_2 root system in W , Cartan matrix $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$; (L5) isospin asymmetry $\Delta n = n_d - n_u = 4 > 0$ (D47, theorem of C4) selects $\mathbf{3}$ over $\bar{\mathbf{3}}$, identifying the matter sector from the axioms. Exact identity: $k_3/(k_1 + k_2) = R_e = 6$. Negative result documented: bijection $(T_2) \leftrightarrow (T_3)$ physically motivated but not S_4 -equivariant. OP-D58-1 and OP-D58-3 resolved.
- Gauge boson masses have been partially derived: the ratio $M_Z/M_W = 1/\cos(19\pi/119)$ is an unconditional theorem (D62); the electroweak vev $v = N_{\text{tot}}^2/(R_e k_1) \cdot m_p$ is conjectured at 676 ppm; and the Higgs mass is predicted as $M_H/m_p = 133.611$ at 1.49σ (D62). Five new open problems (OP-D62-1 through OP-D62-5) are identified.
- The quark mass spectrum has been addressed by two conjectures of distinct ontological character (D63): H_{mass} derives $m_d/m_u = 2401/1104$ as the ratio of coherence-pulsation costs of the valence cores K_{24} and K_{28} , giving $m_u = 2.155$ MeV and $m_d = 4.687$ MeV within 0.37% of PDG values with no additional free parameter; H_{sea} reproduces the sea quark masses m_s, m_c, m_b, m_t within 2.5% of PDG via a residual-fluidity weighting $R_{\text{sea}}/(R_{\text{sea}} + r(n))$. A structural identity $n_u - 1 = p_{k_1} = 23$ connects the hadronic and cosmological sectors for the first time.
- **New in v30 — D64** establishes, for the first time explicitly, a structural correspondence between the surface-degeneracy theorem of D50 ($\Omega_{\text{surf}} = 4^{R_{\text{surf}}}$ microstates on the proton’s active

surface at fixed global quantum numbers) and the soft-hair programme of Hawking, Perry and Strominger, in which a black hole carries an infinite-dimensional space of supertranslation and superrotation charges at fixed mass, charge and angular momentum [11]. The correspondence is presented as a structural analogy, not a claim of mathematical identity. D64 also reports an honestly negative result: a naive nucleon-counting extrapolation of the PDL per-relation degeneracy fails the Bekenstein–Hawking target by 36.7 orders of magnitude, a discrepancy proved mass-independent by algebraic construction and verified identically across four black holes spanning eight orders of magnitude in mass (one solar mass, M87*, Sagittarius A*, GW150914 remnant). Two open problems are introduced: OP-D64-1 (the genuine microstate-counting problem for macroscopic assemblies) and OP-D64-2 (a fully relational unit system deriving c and μ^* from C1–C4).

- **New in v30 — D65** is a direct sequel to D64, asking whether PDL predicts shared *behaviours* between its two most extreme closures — the isolated proton and the macroscopic black hole horizon — independently of any absolute microstate count [12]. Five results are established: (1) any closed spherical PDL surface is forced, by a combinatorial analogue of the Poincaré–Hopf theorem verified independently by spinor relaxation, exact discrete Gauss–Bonnet computation, and holonomy calculation, to carry exactly two coherence defects, never zero — a structural, axiom-level account of why no black hole is ever observed with exactly zero spin; (2) the curvature-migration exponent governing the approach to extremality is exactly 1, once floating-point noise is eliminated by arbitrary-precision arithmetic, in tension with the third law of black hole thermodynamics, reported openly as OP-D65-2; (3) a dissolution threshold derived from the nucleon saturation number $Z_{\text{sat}} = 20$ (D22) gives a critical mass $M^* \approx 4.3 M_\odot$ falling inside the observed compact-object mass gap, an output rather than an input of the construction; (4) seven independent attempts at an absolute microstate count or a Bardeen–Thorne-type spin-limiting mechanism are reported as failures, each with its precise structural cause; (5) these results are organised under the *behaviour-versus-magnitude principle*: PDL currently supports claims about what is topologically forced or what functional form a law must take at the black hole scale, but not yet about absolute magnitude, and this asymmetry is argued to be structural rather than a temporary shortage of derivation effort. Three open problems are introduced (OP-D65-1 through OP-D65-3), the third of which — constructing the coherence stress-energy tensor and equation of state for an inhomogeneous, symmetry-broken closure rather than the homogeneous limit — is identified as the single most consequential open problem for future PDL work on black holes.
- **New in v30 — PDL–OFN collaboration, Session 67.** An in-depth exchange with Oleg Evdokimov produced five precise connections between PDL and the Ontological Fundamental Network (OFN), reported here for the first time. (i) The tripartition $\{A, B, C\}$ of $K_{2,2,2} = L(K_4)$ is exactly the set of the three perfect matchings of K_4 (D58 Lemma L2, D61) — the first PDL–OFN correspondence with status *mathematical identity* rather than candidate analogy. (ii) The algebraic identity $\varphi + \gamma/2 = 2$ holds exactly in $\mathbb{Q}(\sqrt{5})$, where φ is the PDL golden ratio and $\gamma = 3 - \sqrt{5}$ is the OFN spectral gap of Ω_{21} ; the filter factor $k = [R\varphi]^{1/18}$ (with R rational) lies in $\mathbb{Q}(\sqrt{5})^{1/18}$, not in $\mathbb{Q}(\sqrt{5})$ itself. (iii) $L(K_4) \cong K_{2,2,2}$ is verified computationally; a claimed application of Whitney’s theorem is corrected, since Whitney’s isomorphism theorem does not apply to K_3 and K_4 (exceptional cases): $\text{Aut}(K_{2,2,2}) = S_2 \wr S_3$, of order 48 (the octahedral group O_h), not S_4 of order 24. (iv) K_4 has 4 vertices and 6 edges, $4 + 6 = 10$, matching $\dim \mathcal{P}(1, 3) = 4$ translations + 6 Lorentz generators = 10 (noted already in D35/D61). (v) A reinforced lockdown script for the identity $n_u - 1 = p_{k_1} = 23$ (D63) returns 13/13 checks passed, with isolation at 0.265% against 378 pre-registered candidate pairs. Several proposals were evaluated and correctly identified as post-hoc numerology or algebraically redundant, and are documented as negative results with the same rigour as positive ones, following the protocol of Section 1.3. Note N02, *From $\mathbb{Z}_3$ to Three Generations: A PDL–OFN Bridge*, is in preparation (PDL side drafted and sent to Oleg Evdokimov; the OFN-side section and final deposit are pending).

There is no longer any foundational open problem at the axiomatic level. The causal chain from C1–C4 to the cosmological constant Λ_{PDL} is complete. The equation of state of the coherence fluid (D54) is fully resolved. The Weinberg angle (D55) is derived as an unconditional theorem, closing OP10 at the level of θ_W . The remaining open problems concern higher-level derivations (electroweak W/Z mass ratio, quark mass spectrum, three fermion generations), extensions (superheavy nuclei, dark matter identification, macroscopic black hole entropy counting), and experimental confrontations. A

new structural hypothesis OP-D59-2 is identified: the minimal covariant derivative $D_\mu = \partial_\mu - igA_\mu$ may be the unique C4-compatible gauge transport, making gauge dynamics a consequence of the survival principle. At the black hole scale specifically, the programme has reached a stable and honestly delimited position: behavioural predictions (D64, D65) are established with the same rigour as the rest of the corpus, while the absolute microstate count remains open (OP-D64-1) pending the identification of a $K_4 \leftrightarrow K_4$ coupling rule for macroscopically many engaged nucleons — the single most consequential missing piece for extending the causal chain beyond a single isolated closure.

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1 Purpose and Scope

1.1 Aim of this document

This mapping document pursues four objectives. First, it provides a complete and current inventory of the PDL corpus (D01–D48, plus auxiliary documents), with concise descriptions of each document’s role and main result. Second, it makes explicit the logical dependency structure of the programme: which results follow from which, what has been proved, and what remains conjectural or open. Third, it classifies all results according to a rigorous epistemic taxonomy, distinguishing proved theorems, conditional theorems whose hypotheses are explicitly named, structurally motivated conjectures, and open problems. Fourth, it identifies the most productive entry points and continuation paths for a colleague wishing to extend the programme. This document is not a replacement for the primary technical documents; it is a navigational and epistemic guide that complements them.

1.2 What makes this version different from v29

Version 29 covered the corpus up to D63 and established the gauge boson mass ratio $M_Z/M_W = 1/\cos(19\pi/119)$ (D62) and the two quark-mass conjectures H_{mass} and H_{sea} (D63). The present version v30 adds Documents D64 and D65, extending the programme for the first time to a systematic, honestly delimited treatment of black hole physics beyond the Bekenstein–Hawking area law and PBH threshold of D37/D38/D45; it also records the outcome of Session 67 of the PDL–OFN collaboration with Oleg Evdokimov, and the current status of Note N02.

- **New in v30 — D64** establishes the soft-hair correspondence between the PDL surface-degeneracy theorem (D50) and the Hawking–Perry–Strominger soft-hair programme, as a structural analogy rather than a mathematical identity. It reports, with full rigour, a negative result (the naive nucleon-counting extrapolation fails the Bekenstein–Hawking target by 36.7 orders of magnitude, mass-independently) and introduces OP-D64-1 and OP-D64-2.
- **New in v30 — D65** is a direct sequel to D64. It shifts the question from absolute microstate counting to shared *behaviours* between the proton-scale and black-hole-scale closures: the forced two-coherence-defect theorem (never zero spin), the curvature-migration exponent 1 near extremality (a documented open tension with the third law), the nucleon-dissolution mass threshold $M^* \approx 4.3 M_\odot$, and the behaviour-versus-magnitude principle that organises seven further negative results. It introduces OP-D65-1 through OP-D65-3.
- **New in v30 — PDL–OFN Session 67.** Five precise PDL–OFN connections are recorded (Section 14): the mathematical identity between the $K_{2,2,2} = L(K_4)$ tripartition and the perfect matchings of K_4 ; the exact algebraic identity $\varphi + \gamma/2 = 2$ in $\mathbb{Q}(\sqrt{5})$; the corrected automorphism group $\text{Aut}(K_{2,2,2}) = S_2 \wr S_3$ (Whitney’s theorem does not apply to K_4); the $4 + 6 = 10$ structural echo of $\dim \mathcal{P}(1, 3)$; and a reinforced lockdown script for $n_u - 1 = p_{k_1} = 23$ (13/13 PASS). Note N02 (*From $\mathbb{Z}_3$ to Three Generations*) is in preparation; the PDL-side draft is complete and awaits the OFN-side contribution.
- A new Section 7, “Black Hole Physics II: Soft Hair, Behavioural Universality, and the Microstate-Counting Problem”, is added between the Black Hole Thermodynamics section and the Coherence Stress-Energy Tensor section, together with a new dependency figure (Figure 3).

The reader familiar with v29 should note: D64 and D65 added to the corpus table and bibliography; five new open problems added (OP-D64-1, OP-D64-2, OP-D65-1, OP-D65-2, OP-D65-3); the epistemic table gains a new layer (L18, black hole physics II); the falsifiable predictions table gains the mass-gap output $M^* \approx 4.3 M_\odot$; the continuation guide and glossary are updated accordingly; the $K_4 \leftrightarrow K_4$ coupling rule for macroscopically many engaged nucleons is identified as the single highest-priority entry point for the continuation of the black hole physics programme.

1.3 Epistemic conventions

Throughout this document, the following colour-coded epistemic labels are used in tables and text:

Theorem A result established by a complete proof from the PDL axioms C1–C4 and previously proved theorems, with no free parameters and no unproved auxiliary hypotheses.

Theorem (Hx) A result proved under one or more explicitly named hypotheses $H1, H2, H3, \dots$; the proof itself is complete conditional on those hypotheses.

Conjecture A structurally motivated claim supported by numerical evidence, qualitative arguments, or partial proofs, but not yet established rigorously.

Named Hypothesis An auxiliary assumption that is explicitly labelled and isolated, so that results depending on it are clearly traceable.

Open Problem A well-posed question whose resolution would advance the programme significantly.

2 Document Corpus

2.1 Complete document table

Table 1 lists the complete PDL corpus in Zenodo canonical order.

Table 1 PDL corpus in Zenodo canonical order. *Main result* denotes the single most consequential contribution of each document to the programme chain.

Label	Title (abbreviated)	Main result / role
D01	The Emergence of Physical Reality within the PDL Framework [13]	Founding article: four relational axioms, structure, proton architecture, first treatment of α, μ, G , and cosmology.
D02	Introduction to the PDL [14]	High-level conceptual overview; standard first reading for new collaborators.
D01F	L'émergence de la réalité physique (LDP FR) [15]	French translation of D01.
D03	The Emergence of Physical Reality (IMRaD format) [16]	Journal-formatted version of D01.
D04	A Modal Dialogue between PDL and the Theory of Objectivity [17]	Comparative study of PDL and TO; ontological commitments of PDL.
D05	A Minimal Relational Sketch for the Golden Ratio in PDL [18]	Self-similar core–surface partition; golden ratio and active surface $(\sigma, 3)$.
D06	Hierarchical Filtering of Coherence Leakage and the Exponent 18 [19]	First formulation of the 18-filter leakage hierarchy.

Label	Title (abbreviated)	Main result / role
D07	Gleason-Type Uniqueness of Born's Rule for Spin- $\frac{1}{2}$ [20]	Early sketch of the Gleason approach; superseded by D34 at Level 1.
D08	Logical Leakage as Self-Maintained Probability: Topological Reformulation [21]	Coexistence topology; coherence-cost pseudo-metric; leakage as probability.
D09	PDL as a Foundational Research Programme [22]	Programme position paper; first systematic open-problem list.
D10	From Discrete Coherence Flux to Effective Fields [23]	Gauss- and Faraday-like relations from coherence fluxes; path towards Schrödinger.
D10a	Proper Time as Coherence-Cycle Counting [24]	Proper time as a count of coherence cycles; emergent Minkowski structure; $\omega_p = m_p c^2 / \hbar$ forces $\lambda_C = \hbar / (m_p c)$ and $V_C = (4/3)\pi\lambda_C^3$ (chain 1 confirmed in D48v3; chain 2 is a geometric consistency check, not an independent proof — see D48v3 Remark 3.1).
D11	Towards an Einstein–Dirac Unification from PDL [25]	Preliminary sketch coupling Dirac spinors to the emergent coherence-cost metric.
D12	A Structural Derivation of the Fine-Structure Constant from PDL [26]	Closed-form expression for α from $(930, \sigma, \varphi)$; accuracy $\mathcal{O}(10^{-4})$.
D13	Compatibility of the Schrödinger Framework with PDL [27]	Benchmark: PDL-derived α and μ in hydrogenic Schrödinger; spectral concordance.
D14	Born's Rule and the Golden-Ratio Active Surface in PDL [28]	Relational derivation of Born's rule for spin- $\frac{1}{2}$; role of φ in σ .
D15	Towards a Schrödinger-Type Dynamics from PDL: A Relational Sketch [29]	Early sketch of Schrödinger emergence; superseded by D32.
D16	Combinatorial Selection of the Proton Architecture: Uniqueness Conjecture [30]	Optimisation functional for the quintuplet; local uniqueness conjecture.
D16a	Minimal Stationary Closures in PDL: Necessity of the $(4, 6)$ Block [31]	Theorem: no admissible closure exists on $ V < 3$; K_4 is first admissible.
D16b	Combinatorial Selection and Local Uniqueness of the Proton Architecture [32]	Refined combinatorial treatment; local uniqueness conjecture for $(24, 28, 930, 10087, 11017)$.
D17	Coherence Leakage, Hierarchical Filtering, and the Exponent 18 [33]	Organisation of the 18 filters into three blocks; effective coupling ansatz ε^{18} .

Label	Title (abbreviated)	Main result / role
D18	Discrete Cavity Modes and Logical Density of States [34]	PDL cavity modes; $2n^2$ density-of-states in three dimensions.
D19	Existence as Pulsating Closure: An Ontological Meditation [35]	Philosophical analysis of PDL boundaries, leakage, and transcendence.
D20F	Qui que nous puissions être (FR) [36]	French philosophical synthesis; companion to D20.
D20	Whoever We May Be: A Philosophical Synthesis of PDL [37]	Ontological and existential synthesis; accessible entry point for non-specialists.
D21	Universal Coherence Leakage: A Structural Bridge between G and α [38]	Primary result: $R_{\text{tot}}/R_{\text{surf}} = 2\nu^2/(2\nu^2 - 1)$; $G \approx 6.67448 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$ (27 ppm vs. CODATA 2022).
DN	<i>Whatever We May Be</i> : The PDL popular book [39]	Popular science monograph; non-technical introduction to PDL.
D22	Nuclear Stability and the Periodic Table as Combinatorial Closure Hierarchies [40]	Neutron quintuplet (24, 28, 1032, 9960, 10992); $N_{\text{crit,max}} = 126.1$; $Z_{\text{sat}} = 20$; Mirror Lemma (confirmed analytically in D47).
DM	PDL Global Mapping v20 [41]	Navigation document for the complete corpus D01–D53; the present document.
D23	Topological Origin of the Exponent 18, S^2 , and the Periodic Table [42]	Theorem: $18 = 6 + 5 + 4 + 3$ is topologically necessary via S^2 ; $n = 3$ forced by S^2 topology (consistency with D48v3; the geometric argument is not an independent chain for V_C — see D48v3 Remark 3.1).
D24	Closure-Density Dependence of κ and the Hubble Tension [43]	Introduction of $\sigma(N) = 1 - (1 - \kappa)^N$; structural origin of H_0 discrepancy.
D25	A Parameter-Free Structural Bridge between α and G [44]	Primary result: full bridge formula; G at 27 ppm; $\kappa \approx 0.0075194$.
D26	Cosmological Resolution via PDL: G as Topology-Dependent [45]	G as topology-dependent parameter; Hubble tension as scale-transition signature.
D27	Structural Derivation of N_{CMB} from the PDL Neutron Architecture [46]	$N_{\text{CMB}} = 40$ from n_6 ; $H_{0,\text{CMB}} = 67.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$; 0.27σ Planck.
D28	The PDL–QCD Boundary: Mass Ratio, Correction, and the ε_G^B Conjecture [47]	$\mu = \frac{155}{11017}(1 - 2\kappa^2) \frac{m_{\text{iso}}}{m_p}$; ε_G^B conjecture formulated.
D29	Axiomatic Derivation of 155/11017: Gate 1 Resolved [48]	Theorem: 155/11017 proved from C1–C4 via AB; 768/768 exhaustive verification.

Label	Title (abbreviated)	Main result / role
D30	QCD Correction Coefficient $a = 2$: Gate 2 Resolved [49]	Theorem: $a = 2$ structurally forced; $m_{\text{iso}} = m_d - m_u$ is the unique irreducible external parameter.
D31	Proof of $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$: Gate 3 (preliminary) [50]	First proof of Gate 3 under linear bridge inversion hypothesis; superseded in rigour by D36.
D32	Schrödinger Dynamics from the AB Coupling Criterion [51]	Theorem: Schrödinger equation derived; $b = -i\hbar/(2m_e)$; ∂_x^2 forced by AB.
D33	Dirac Equation from the $\text{SL}(2, \mathbb{C})$ Pulsation of K_4 [52]	Theorem: $T = -i\sigma_2$ unique among 8 candidates; $T^2 = -I_2$; Clifford algebra; NR limit = D32 exactly.
D34	Born Rule from AB-Admissible Amplitudes [53]	Theorem Level 1: five Gleason axioms verified; Born rule unique from Gleason uniqueness.
D35	Quantitative Einstein Equation from $\sigma(N)$ [54]	Theorem (now unconditional via D42): $G_{\text{eff}}(N) = G_{\text{Newton}}$; Hubble ratio at 0.006%; synthesis document.
D36	Proof of $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ from the Trace Structure of K_4 [55]	Theorem (now unconditional via D42): $\sigma_0 = 0$ exact from AB; $\text{Tr}(A) = 0$ proved; Gate 3 strengthened.
D37	Surface Locality of Relational Entropy and the PDL Area Law [56]	Theorem BH-1: entropy of a PDL closure is entirely carried by R_{surf} ; area law $S \propto R_{\text{surf}}$ unconditional.
D38	Bekenstein–Hawking Entropy from the PDL Effective Gravitational Coupling [57]	Theorem BH-2 (now unconditional via D42): $S_{\text{BH}}/k_B = 4\pi(M_{\text{eff}}/M_{\text{Pl}})^2$; verified to 10^{-15} ; three falsifiable PBH predictions.
D39	Derivation of κ from the PDL Axioms: the Indifference Lemma [58]	Theorem (H3): $\kappa = 3/\sigma$ under H3; partial resolution of OP1; D36 $\Rightarrow$ H3 is the unique residual; H3 now a theorem (D42).
D40	Nuclear Stability and the Periodic Table from PDL Combinatorial Axioms [59]	Theorems: $N_{\min}(Z \leq 20) = Z$ exact; $C(Z > 20) = 190 T_{pp}$ exact; valley of stability at 100% for $Z = 1\text{--}82$; eight assembly rules; connection identified between OP14 and OP12.
D41	Structural Closure Multiplicity and the Isospin-Symmetric Island of Inversion [60]	First PDL confrontation with nuclear spectroscopy data (Ha et al. 2025). Three independent observables consistent with $T = (\Delta n + 1)^2 = 25$: BE2 ratio (22%); N_{comp} ratio (2%); ΔS_{WF} (25%). Conjecture H_B introduced; falsifiable predictions P7 and P8.
D42	Derivation of H3 from Axioms C1–C4: The Equiparticipation Lemma [61]	Theorem: H3 derived unconditionally via D42-L1 (Equiparticipation), D42-L2 (cross-triangle blindness), D42-L3 (S_4 -equivariance; 24,576 cases). Corollary: $\kappa = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$ unconditional. OP1 fully resolved.

Label	Title (abbreviated)	Main result / role
D43	Numerical Verification of the PDL Causal Chain: 18-Filter Architecture and Colab Infrastructure [62]	Systematic numerical verification at 50-decimal precision. Proves $\varepsilon_{\text{geom}}(p) = 329/10087$ and $\varepsilon_{\text{geom}}(n) = 468/9960$ as unconditional theorems (OP-A resolved). Reusable Colab infrastructure.
D44	Closure of OP-B: Derivation of the Hierarchical Filter Factor k from the PDL Axioms [63]	Theorem: $k = 0.921716$ derived from C1–C4 unconditionally. OP-B resolved; OP8 resolved; complete causal chain C1–C4 $\rightarrow G$ closed.
D45	Primordial Black Hole Evaporation Threshold from PDL: A Falsifiable Prediction for Fermi-LAT [64]	First falsifiable astrophysical confrontation. $M_{\text{PDL}}^* = \sigma(40)^{-2/3} M_{\text{GR}}^* \approx 5.706 \times 10^{14} \text{ g}$ (+11.89%). Excess Hawking flux at $E_{\text{peak}} \approx 90\text{--}104 \text{ MeV}$ in IGRB; test protocol via BLACKHAWK + ISATIS + Fermi-LAT.
D46	Born’s Rule Level 2: Derivation of U(1) Phase Freedom from the K_4 Pulsation [65]	Theorem OP4 resolved: Proposition 1 — K_4 sign equivalence $s \sim -s$ and $T^2 = -I_2$ generate a canonical U(1) action with Hopf fibration $S^1 \hookrightarrow S^3 \rightarrow S^2$. Proposition 2 — $(A) \wedge (B)$ criterion is U(1)-invariant (verified over all 8 configurations). Proposition 3 — unique U(1)-equivariant measure on S^3 satisfying Level 1 Gleason axioms is Born’s rule. Quantum dynamics layer complete. Connection to Berry phase and London/Meissner analogy via $C^{(\text{orb})}$ (D48) documented.
D47	Derivation of Sub-Shell Filling Rates and the Periodic Table from the PDL Axioms: Resolution of OP13 and OP14 [66]	Theorem OP13 resolved: $\Delta n = 4$ is the unique value in $\{0, 4, 8, \dots\}$ such that the discriminant of the quasi-completeness equation is a perfect square (149^2), forcing $n_u = 24$, $n_d = 28$, $s = 1/12$ unconditionally. Mirror Lemma (Lemma 5.2): proton and neutron HO-PDL levels are isomorphic, 20 levels verified. Theorem OP14 resolved: $r_{\text{exc}}(Z) = 0$ for $Z \in \{28, 50, 82\}$ and $r_{\text{exc}}(Z) = 1$ otherwise, 31/31 sub-shell boundaries, zero free parameter. Corollary: $N_{\text{min}}(Z)$ for $Z = 1\text{--}82$ is an unconditional theorem of C1–C4. Nuclear stability layer complete.
D48v3	Derivation of the Coherence Stress-Energy Tensor from the PDL Axioms: Explicit Form of C_{coh} , Resolution of OP2-D35 and OP-spin [67]	Theorem OP2-D35 and OP-spin fully resolved (D48v3, doi 10.5281/zenodo.20151380). V_C theorem via D10a (Chain 1); $\rho_{\text{coh}}(N)$ unconditional. $C^{(\text{mass})}$ unconditional theorem. $C^{(\text{spin})} = 0$ in the ground state: theorem from $\langle S^i \rangle = 0$ and $\langle S^i S^j \rangle = \frac{1}{4} \delta^{ij}$ over all 8 K_4 configurations (Weyssenhoff–Raabe). $C^{(\text{orb})} = m_p^2 \langle j_\mu j_\nu \rangle / \rho_{\text{coh}}$ (Madelung), vanishes in ground state (D49). $C^{(\text{leak})}$ in J m^{-3} via D51+D52. PDL Einstein equation dimensionally consistent. Four corrections from v1/v2 documented (Section 10 of D48v3).

Label	Title (abbreviated)	Main result / role
D49	Derivation of the London Equation from the PDL Axioms: Resolution of OP-London [68]	Theorem OP-London resolved. Lemma: $\nu = \frac{1}{4} \ \psi_1 - T^k \psi_1\ _{\mathcal{H}_{\text{cyc}}}^2$ verified 768/768 in exact integer arithmetic. Axiom C4 forces $\psi_j = \psi_1$ over V_C (via C3 connectivity), hence $\phi = \text{const}$ (London gauge). Corollary: $j_s^i = -(n_{\text{coh}} e^2 / m_p c) A^i$ with $n_{\text{coh}} = \sigma(N) R_{\text{surf}} / (V_C R_{\text{tot}}) \in \mathbb{Q}(\sqrt{5})$; $\lambda_{L,\text{PDL}}(N = 40) \approx 7.25 \times 10^{-15} \text{ m}$.
D50	Derivation of the Bekenstein–Hawking One-Quarter Coefficient from the PDL Axioms: Resolution of OP12/BH-3 [69]	Theorem OP12/BH-3 resolved. The coefficient $\frac{1}{4}$ in $S_{\text{BH}} = k_B A / (4\ell_P^2)$ equals the stable fraction $4/16 = 1/4$ per surface relation, proved from D29 (exhaustive, 768/768) and H3 (D42). Statistical independence $P(e_1 \wedge e_2) = 1/16 = (1/4)^2$ verified over 30 720 cases. $\Omega_{\text{surf}} = 4^{R_{\text{surf}}}$; $S_{\text{surf}} = k_B R_{\text{surf}} \ln 4 = 695.352 \text{ nats}$. Black hole thermodynamics layer complete.
D51	The Cosmological Leakage Constant from the PDL Axioms: A Structural Derivation of C and Resolution of OP1-D35 [70]	Theorem OP1-D35 structurally resolved. $\beta_1(K_4) = 3$ gives three leakage cycles (topology). C1+C3 forces prime cycle lengths. Indices $k_1 = n_d - n_u + R_e - 1 = 9$, $k_2 = n_u - R_e + 1 = 19$, $k_3 = R_e \cdot n_d = 168$ are exact theorems; $k_1 + k_2 = n_d$. Conjecture PDL-C: $C = (1 - \kappa)^{997} \times (930/11017)^{23} \times (1 - \eta_L)^{67}$ at 0.17 ppm of Λ_{obs} . Uniqueness verified by exhaustive exclusion of all alternative bases.
D52	Formal Identification of the Three Leakage Bases as Natural Representatives of the Three Cycles of $\beta_1(K_4) = 3$: Completion of Conjecture PDL-C [71]	Theorem OP1-D35 fully resolved. The three bases $(1 - \kappa)$, $R_{\text{val}}/R_{\text{tot}}$, $(1 - \eta_L)$ are proved to be the natural and unique representatives of the three leakage cycles: $(1 - \kappa)$ from D42 (surface sector), $R_{\text{val}}/R_{\text{tot}}$ from D29 Lemma 2 ($\Omega_{\text{val}} = 1$, valence sector), $(1 - \eta_L)$ from D17/D30 (coupling sector). Algebraic independence verified (40 401 cases). Full causal chain: $\text{C1–C4} \rightarrow K_4 \rightarrow \beta_1 = 3 \rightarrow (p_{k_1}, p_{k_2}, p_{k_3}) \rightarrow C \rightarrow \Lambda_{\text{PDL}}$. Cosmological leakage layer complete.
D53	From Four Axioms to the Cosmological Constant: Causal Closure of the Chain C1–C4 to Λ — A Synthesis and Computational Verification [72]	Synthesis and verification of the complete chain $\text{C1–C4} \rightarrow \Lambda_{\text{PDL}}$ in a single self-contained document. Independent numerical verification at 50-decimal precision: C_{PDL} confirmed to 0.41 ppm relative to D51; prime robustness confirmed (composite substitution degrades agreement by ≥ 4 orders of magnitude); metrological origin of the apparent 8.17 ppm offset identified (8.6 ppm in $(m_p c / \hbar)^2$ between CODATA versions, no structural significance). The chain spans 57 orders of magnitude.

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D54	Equation of State of the PDL Coherence Fluid: Resolution of OP-pressure [1]	Theorem OP-pressure resolved. The equation of state $w(N) = -\rho_\Lambda/\rho_{\text{coh}}(N)$ is an unconditional theorem of C1–C4. In the homogeneous ground state: $w_{\text{mass}} = 0$ (pressureless dust, D48v3 Thm 4.1); $w_{\text{spin}} = 0$ (D48v3 Thm 5.3, OP-spin); $w_{\text{orb}} = 0$ (D49 Thm 1 via London gauge); $w_{\text{leak}} = -1$ (cosmological constant structure, D48v3 Prop. 8.1). PDL dark-energy fraction $\Omega_\Lambda^{\text{PDL}} = 0.6838$ vs $\Omega_\Lambda^{\text{obs}} = 0.685$ (Planck 2020): agreement to 0.17%, no cosmological parameter as input. The PDL coherence fluid provides a unified description of pressureless cold matter (at sub-nuclear scales, $ w \sim 10^{-44}$) and dark energy (at cosmological scales) from a single combinatorial structure. Two new open problems introduced: OP-DM (dark matter identification) and OP-inhomogeneous (equation of state beyond the ground state).
D55	Derivation of the Weinberg Angle from the PDL Axioms: De-phasing Sequencing, Lemma D, and a New Prediction for Δm_{iso} [2]	Theorem OP10 partially resolved (θ_W derived). Lemma D (new unconditional theorem): the $(A) \wedge (B)$ -stable interface decomposes into $N_{\text{total}} = (n_d/4) \cdot (r_d - r_u)/R_e = 7 \times 17 = 119$ elementary steps, proved exhaustively over 768 configurations with zero counter-examples. Main result: $\theta_W = k_2\pi/119 = 19\pi/119$, where $k_2 = n_u - R_e + 1 = 19$ (D51 theorem). Berry phase identification: $\gamma_B = -\Omega/2$ with $\Omega = 4\pi k_2/119$ (D46 + Lemma D). Predicted $\sin^2 \theta_W = 0.231196$ vs PDG 2024: 0.23121 ± 0.00003 , compatible at 0.48σ . New prediction P10: $\Delta m_{\text{iso}} = 2.446$ MeV (within 0.92σ of FLAG 2024). The electroweak sector gauge geometry is structurally derived from C1–C4. Remaining open: W/Z mass ratio (OP10-c) and lattice QCD test (OP10-d).
D56	Derivation of the Linear Scaling $N_{\text{comp}}(k) = k$ from Axioms C1–C4: Resolution of OP-D41-1-A [3]	Theorem OP-D41-1-A resolved. Three lemmas: L1 (uniqueness on one unit, D16a), L2 (disjointness of mixed-triangle sets, D29), L3 (C3-irreducibility). $N_{\text{comp}}(k) = k$ verified for $k = 1, \dots, 5$.
D57	Derivation of the Tree-Level Weinberg Angle and the SU(2) Gauge Structure from C1–C4 [6]	Theorem : $\sin^2 \theta_W^{(\text{tree})} = \sin^2(\pi/R_e) = 1/4$, two independent routes (Route A: $192/768$, D29; Route B: $\dim(\mathbf{1})/\dim(\text{orbit-4})$), equivalence via $ V_4 = 4$. Conjecture H_{SU2} introduced (resolved by D60 as unconditional theorem). New: $A_4/V_4 \cong \mathbb{Z}_3 \leftrightarrow \beta_1(K_4) = 3 = b_1(\Omega_{21})$ (partial resolution OP-OFN-1). D55 reinterpreted as Berry correction of π/R_e : $\theta_W = (114/119) \cdot (\pi/6)$. DOI 10.5281/zenodo.20600264.

Label	Title (abbreviated)	Main result / role
N01	PDL–OFN Convergence on $\beta_1 = 3$: A Numerical Investigation (Laubscher & Evdokimov) [7]	Five Python scripts verify exhaustively: $\beta_1(K_4) = 3$ (PDL) = $b_1(\Omega_{21}) = 3$ (OFN, Hamming dist-1 adjacency). Poses OP-OFN-1 (formal link 3 PDL cycles $\leftrightarrow$ 3 OFN generations) and OP-OFN-4 (derive $SU(3) \times SU(2) \times U(1)$ using OFN decomposition $8 \oplus 3 \oplus 1 \oplus 1$ as guide). Partial resolution of OP-OFN-1 via $A_4/V_4 \cong \mathbb{Z}_3$ (D57).
N02	<i>In preparation.</i> From $\mathbb{Z}_3$ to Three Generations: A PDL–OFN Bridge (Laubscher & Evdokimov) [73]	PDL-side draft complete (B2_PDL_OFN_bridge.tex); OFN-side section and final co-authored deposit pending. Records the five Session-67 connections of Section 14, in particular the mathematical identity between the $K_{2,2,2} = L(K_4)$ tripartition and the perfect matchings of K_4 , and the reinforced lockdown script for $n_u - 1 = p_{k_1} = 23$ (13/13 PASS). Not yet in the Zenodo DOI registry.
D58-py	PDL_SU3_script1.py: Verification of the Five Lemmas Establishing $SU(3)$ as an Unconditional Theorem of C1–C4 [74]	Supplementary verification script for D58. Five lemmas verified exhaustively by exact integer arithmetic (Python, numpy, networkx). doi 10.5281/zenodo.20623231. Deposited as Software on Zenodo.
D58	Derivation of the $SU(3)$ Gauge Structure from the PDL Axioms C1–C4: Completion of the Standard Model Gauge Group [75]	Theorem (algebraic): $SU(3)$ from C1–C4 + Cartan–Killing–Weyl. Five lemmas (L1: $S_4/V_4 \cong S_3$; L2: S_4 -equivariant bijection $V_4 \setminus \{e\} \leftrightarrow V_4$ -edge-orbits; L3: $A_4/V_4 \cong \mathbb{Z}_3 = Z(SU(3))$; L4: rank-2 Cartan from $R_e = 6$; L5: A_2 root system). Negative result: V_4 acts faithfully on $H_1(K_4; \mathbb{R})$. Corollary: $SU(3) \times SU(2) \times U(1)$ algebraic theorem of C1–C4 (with D46, D57). doi 10.5281/zenodo.20622987.
D59-py	PDL_D59_script1.py: Verification of the Five Lemmas Establishing the Fundamental Representation $\mathbf{3}$ of $SU(3)_c$ from C1–C4 [10]	Supplementary verification script for D59. Five lemmas (L1–L5) and one negative result (NEG) verified exhaustively by exact integer arithmetic (Python standard library only: itertools, fractions; no floating-point). All 17 checks PASSED OR VERIFIED NEGATIVE. doi 10.5281/zenodo.20628926. Deposited as Software on Zenodo.
D59	Physical Identification of $SU(3)_c$ and the Fundamental Representation $\mathbf{3}$ from the PDL Axioms C1–C4 [76]	Theorem : $SU(3)_c$ acts on the trace-zero plane $W = \{a + b + c = 0\} \subset \mathbb{C}^3$ in the fundamental representation $\mathbf{3}$, as an unconditional theorem of C1–C4 + Cartan–Killing–Weyl. Axes of W labelled by $V_4 \setminus \{e\}$ (D58, L2). Orientation $\mathbf{3}$ vs $\bar{\mathbf{3}}$ selected by $\Delta n = 4 > 0$ (D47). Exact identity $k_3/(k_1 + k_2) = R_e = 6$. Negative result: bijection $(T_2) \leftrightarrow (T_3)$ not S_4 -equivariant. Resolves OP-D58-1 and OP-D58-3. doi 10.5281/zenodo.20629282.

Label	Title (abbreviated)	Main result / role
D60	From the Logic of Existence to the Gauge Symmetry Group: A Proof of Hypothesis H_{SU2} from Axioms C1 and C2 [5]	Theorem OP-D57-1 resolved: H_{SU2} is an unconditional theorem of C1+C2 alone. The V_4 -invariance of the pulsation is forced by the co-originary of $s^{(1)}$ and $s^{(2)}$ (C1): $s \rightarrow -s$ breaks C2 (negative result, 8/8 verified); orbit-4 $\cong V_4$ as regular set; pairings = cosets of H_1, H_2, H_3 ; C1-admissibility $\Leftrightarrow V_4$ -invariance on all of $Coh(K_4)$; $G_{\text{eff}} = S_4/V_4 \cong S_3$. Two verification scripts (PDL_C1V4_script1.py, PDL_C1V4_script2.py) deposited with D60. Corollary: $SU(3) \times SU(2) \times U(1)$ + representation 3 : no conjectural step remaining in the gauge sector. DOI 10.5281/zenodo.20639684.
D61	Gauge Dynamics as a C4 Survival Theorem: Derivation of the Minimal Covariant Derivative from the PDL Axioms C1–C4 [4]	Theorem OP-D59-2 resolved: $D_\mu = \partial_\mu - igA_\mu$ derived from C4 via four arrows. (α) C4-admissible transports = $Coh(K_4) \cong \mathbb{Z}_2^3$ (2048 exhaustive checks). (β) V_4 acts faithfully on $Coh(K_4)$ by automorphisms; fixed-point subgroup $F \cong V_4$ labelled by the three perfect matchings of K_4 = axes of W (D59); semidirect product $\mathbb{Z}_2^3 \rtimes V_4$ order 32 (288 checks). (γ) Consistency with $G_{\text{gauge}} = SU(3) \times SU(2) \times U(1)$ (theorem via D46–D60). (δ) Utiyama (1956) uniqueness: D_μ is the unique first-order equivariant operator; local gauge invariance is a C4 survival condition, not a postulate. Gauge dynamics without variational postulate. Two verification scripts deposited with D61. DOI: to be assigned.
D62	Gauge Boson Masses from Combinatorial Axioms: Derivation of v , M_Z , M_W and Prediction of M_H [77]	Theorem : $M_Z/M_W = 1/\cos(19\pi/119)$ (unconditional, from D55+D57); 4 primary Higgs decompositions in $\mathbb{Z}_2^3$ (unconditional); neutrino exit at cascade level 2 (unconditional). Conjecture T1 : $v = N_{\text{tot}}^2/(R_e k_1) \cdot m_p = 14161/54 \cdot m_p$ [676 ppm PDG]. Conjecture : $M_Z/m_p = 97.179$ via analytical chain $\Delta\alpha(M_Z)$ [84.6 ppm, 3.67 σ ; tension traced to pion/kaon corrections pending D63]. Conjecture : $M_H/m_p = 133.611$ [1.49 σ]. Formal definition of $m_B = \inf\{E > 0 \mid \exists C \in \mathbb{Z}_2^3 : \nu(C) - \nu_0 = E\hbar/(c^2 R_{\text{tot}})\}$, categorically distinct from fermionic closure cost. Five new open problems: OP-D62-1 to OP-D62-5. DOI 10.5281/zenodo.20679631.

Label	Title (abbreviated)	Main result / role
D63	Quark Mass Spectrum from Combinatorial Axioms: Two Conjectures on Valence and Sea Quark Masses [78]	Conjecture H_{mass} (strongly motivated): $m_d/m_u = \mathcal{Q}(K_{n_d}) / \mathcal{Q}(K_{n_u}) = 2401/1104 \approx 2.1748$ [0.59% PDG]; $m_u = 2.155$ MeV [0.22%]; $m_d = 4.687$ MeV [0.37%]. Candidate $f(n) = r(n) \cdot n^3$ isolated by exhaustive scan (unique minimum below 1% in family of degree-≤ 5 monomials). Exact structural identity: $n_u - 1 = p_{k_1} = 23$ (D47 + D51 theorems). Conjecture H_{sea}: $m_{\text{sea}}(n) = [n^4(n-1)/n_u^4(n_u-1)] \cdot [R_{\text{sea}}/(R_{\text{sea}} + r(n))] \cdot m_u$; residual-fluidity weighting selected by exhaustive scan over 5 candidates; m_s [0.45%], m_c [2.46%], m_b [2.37%], m_t [1.97%] all within 2.5% of PDG. Non-hadronisation of top quark: $r(332) = 54946 \gg R_{\text{sea}} = 10087$ (structural consequence). Hadronisation boundary $n_{\text{max}} = 140$, $m_{\text{max}} \approx 7677$ MeV. Lockdown script: 66 checks, 66 PASS. Three new open problems: OP-D63-1 to OP-D63-3. DOI 10.5281/zenodo.20696391.
D64	Soft Hair on PDL Closures: A Structural Correspondence Between Surface Microstate Counting and Black Hole Hair [11]	Structural analogy (not identity) between the D50 surface-degeneracy theorem $\Omega_{\text{surf}} = 4^{R_{\text{surf}}}$ and the Hawking–Perry–Strominger soft-hair programme. Proposition (negative result): naive nucleon-counting extrapolation fails the Bekenstein–Hawking target by 36.7 orders of magnitude, proved mass-independent and verified identically for $M_{\odot}$, M87*, Sagittarius A*, and the GW150914 remnant (four masses, eight orders of magnitude). Surface-restricted count narrows the solar-mass gap to 1.4 orders of magnitude but is shown <i>not</i> to generalise (M87*: 7.9 orders of magnitude, worse). Two new open problems: OP-D64-1 (macroscopic microstate counting), OP-D64-2 (relational derivation of c, μ^*). DOI 10.5281/zenodo.20868328 (v2).

Label	Title (abbreviated)	Main result / role
D65	Two Limits of the Same Surface: Behavioural Correspondences Between Nucleon and Black Hole Closures in PDL [12]	Direct sequel to D64: shifts from absolute microstate counting to shared <i>behaviours</i> . Theorem : any closed spherical PDL surface carries exactly two coherence defects, never zero (combinatorial Poincaré–Hopf, verified by three independent methods). Theorem : curvature-migration exponent = 1 near extremality (arbitrary-precision arithmetic), an open tension with the third law (OP-D65-2). Conjecture : nucleon-dissolution mass threshold $M^* \approx 4.3 M_\odot$ from $Z_{\text{sat}} = 20$ (D22), falling inside the observed compact-object mass gap, untuned. Seven further negative results (rigid-body inertia, two-horizon thermodynamics, geometric curvature saturation, spin stress-energy suppression, a spurious nuclear-magic-number coincidence, polytropic pressure support, spin–horizon-scale correlation). Behaviour-versus-magnitude principle stated as the paper’s central methodological claim. Three new open problems: OP-D65-1 to OP-D65-3. DOI 10.5281/zenodo.21220251.
— PDL-V Programme: Life and Consciousness (DL series) —		
DL01	From Axioms to Life: The PDL-V Conjecture and the Hierarchical Replication of C1–C4 [8]	Computational theorems inaugurating the PDL-V programme. Lifting operator L , constraint-structure operator $S(\Gamma)$, self-representation relation $R^k(\Gamma)$ defined formally. $S(K_4) \cong K_4$: proved over 8/8 coherent configurations. R_{weak}^1 forced at 100% for all $n \geq 4$ (verified $n = 4$ –7, 8/8 to 64/64). $\delta^* = 1/2$: invariant of the optimal lineage variation = half-pulsation of K_4 . Rarity law $f(n) = 2^{n+1-\binom{n}{2}}$ with exponents equal to triangular numbers: exact theorem.
DL02	Existence, Separation, and Finiteness of the Life and Consciousness Thresholds in the PDL-V Programme [9]	Analytical theorems on the life/consciousness hierarchy. Theorem 1: K_4 is the unique fixed point of S — $C(4, 3) = 4$ is the unique positive integer solution of $(n-1)(n-2) = 6$. Theorem 2: $n_{\text{life}}^* < n_{\text{consciousness}}^*$ — structurally forced for $n \geq 5$ from $ S^1 \neq S^2 $. Theorem 3: $n_{\text{max}}^* < \infty$ — from the doubly exponential rarity law. Hierarchy $4 < n_{\text{life}}^* < n_{\text{consciousness}}^* < n_{\text{max}}^* < \infty$: unconditional theorem of C1–C4. Numerical values of thresholds remain open (DL03 programme).

2.2 Reading guide for a new collaborator

The following reading pathways are recommended depending on one’s entry point.

- **Understanding the foundations.** Start with D02 (conceptual overview), then D16a (proof that K_4 is the unique minimal closure), then D01 (full axiomatic development). For the proton architecture, read D16 and D16b for the combinatorial selection arguments.

- **Understanding the constants.** Read D12 for α , then D21 and D25 for the α – G bridge. Read D23 for the topological proof that the exponent 18 is necessary. Then read D29 and D30 for the Gate 1 and Gate 2 proofs. Read D36 for the final Gate 3 proof, and D39/D42 for the full resolution of OP1.
- **Understanding the quantum dynamics.** Read D32 (Schrödinger), D33 (Dirac), D34 (Born Level 1), D35 (Einstein), and D46 (Born Level 2, $U(1)$) in order.
- **Understanding the cosmology.** Read D24, D27, and D35.
- **Understanding black hole thermodynamics.** Read D37 (area law), then D38 (Bekenstein–Hawking and PBH predictions), then D45 (falsifiable confrontation with Fermi-LAT).
- **Understanding the soft-hair correspondence and behavioural black hole physics (new in v30).** Read D64 (soft-hair correspondence, the 36.7-order-of-magnitude negative result, OP-D64-1, OP-D64-2), then D65 (the two-coherence-defect theorem, the curvature-migration exponent, the dissolution mass threshold, and the behaviour-versus-magnitude principle). Section 7 of this document provides a self-contained summary.
- **Understanding nuclear stability and the periodic table.** Read D22, D23, D40, D41, and D47.
- **Understanding the London equation and superconductivity analogy.** Read D46 ($U(1)$ phase freedom), then D49 (London equation from C4 and the metric identity $\nu = \frac{1}{4}\|\psi_1 - T^k\psi_1\|^2$). D50 completes the black hole thermodynamics layer with the one-quarter coefficient.
- **Understanding the cosmological constant.** Read D48 for the leakage term structure, D51 for the derivation of Conjecture PDL-C and the role of prime cycle lengths from $\beta_1(K_4) = 3$, and D52 for the formal identification of the three leakage bases. For a self-contained synthesis of the complete chain with numerical verification, read D53.
- **The closure of the foundational programme.** Read D39 first for context, then D42 for the full proof of H3. After D42, all results previously conditional on H3 are unconditional theorems. Then read D43 (geometric leakage, OP-A) and D44 (filter factor k , OP-B) for the complete causal chain to G . Finally, D51+D52 close the chain to Λ_{PDL} .
- **The coherence stress-energy tensor.** Read D35 for the structural form, then D48 for the explicit component derivation.
- **Numerical verification and the computational infrastructure.** Read D43. It provides self-contained Colab code for every major result and defines the verification protocol for future documents.
- **Continuing the programme.** Read Section 16 for the complete prioritised list of open problems, and Section 17 for the critical-path dependency map.

3 Axiomatic Foundations

3.1 The four relational axioms C1–C4

A logical configuration in PDL is a finite signed graph $G = (V, E, s)$, where V is a finite set of vertices (informational entities), E is a set of undirected edges (binary relations), and $s: E \rightarrow \{1, -1\}$ assigns a sign (agreement or disagreement) to each relation [13, 14]. Four relational axioms are imposed.

Axiom 3.1 (C1 — Minimal binary pulsation). Existence is instantiated as a repeatable, non-trivial alternation between two discrete states, independent of any pre-existing temporal metric. Formally, the discrete dynamical system defined on the space of sign configurations must admit at least one logical 2-cycle: a configuration that returns to itself after exactly two update steps and is not a fixed point.

Axiom 3.2 (C2 — Coherence under logical stress). Every admissible configuration must maintain internal consistency under relational stress. Formally, the graph must be *balanced* in the sense of Harary: every cycle of relations carries a positive product of signs. This is equivalent to requiring that the graph be 2-colourable with respect to signed adjacency.

Axiom 3.3 (C3 — Completeness of relational coverage). Every pair of informational entities must be related, directly or indirectly, by at least one signed relation. Formally, the signed graph G must be connected.

Axiom 3.4 (C4 — Optimisation of coherence cost). Among all configurations satisfying C1–C3, the physically realised configuration minimises the total coherence cost, defined as the fraction of violated triangles (triangles with negative triple product). This is a discrete analogue of a variational principle.

3.2 The minimal admissible closure: K_4

Theorem 3.5 ([31]). *The complete signed graph K_4 on four vertices and six edges is the unique minimal admissible closure of axioms C1–C4: no admissible closure exists on $|V| < 3$ vertices, and K_4 is the first connected balanced complete graph admitting a non-trivial 2-cycle. It is identified with the electron prototype, with relational budget $R_e = 6$.*

3.3 The proton quintuplet

The proton is the minimal hierarchical composite closure admissible under C1–C4, characterised by the integer quintuplet $(n_u, n_d, \sigma, R_{\text{sea}}, R_{\text{tot}}) = (24, 28, 930, 10087, 11017)$, where $\nu_u = 24$ is the number of u -valence cores, $\nu_d = 28$ the number of d -valence cores, $\sigma = 930$ the active surface, $R_{\text{sea}} = 10087$ the sea relational budget, and $R_{\text{tot}} = 11017$ the total relational content [13, 30–32]. The neutron quintuplet is $(24, 28, 1032, 9960, 10992)$ [40].

Conjecture 3.6 (Global uniqueness [30, 32]). The quintuplet $(24, 28, 930, 10087, 11017)$ is the *global* (not merely local) minimum of the coherence leakage functional over all admissible integer tuples. The local uniqueness has been verified computationally for a neighbourhood of radius 10^4 (D43). The global statement remains open (OP2).

3.4 The primary leakage parameter κ

Under the Indifference Lemma H3 (now a theorem, D42) and D39, the primary leakage parameter is:

$$\kappa = \frac{310\varphi}{11017} \in \mathbb{Q}(\sqrt{5}), \quad \varphi = \frac{1 + \sqrt{5}}{2}, \quad \kappa \approx 0.0075194. \quad (1)$$

This is an unconditional theorem of C1–C4 [58, 61]. Under H3 (now a theorem), H1 follows as a theorem [58]: the two-factor decomposition $\kappa = 3/\sigma$ holds. H3 has been derived from C1–C4 in D42 via three lemmas D42-L1, D42-L2, D42-L3; it is no longer a named hypothesis but an unconditional theorem of the programme.

Theorem 3.7 (Gate 3, [55]). *Since H3 is now a theorem [61], H1 follows unconditionally [58], and Gate 3 holds without any named hypothesis: $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ and $\sigma_0 \equiv 0$. The independence of gravitational engagements ($\sigma_0 = 0$ exactly) follows from the AB coupling criterion without additional hypothesis. The tracelessness $\text{Tr}(A) = 0$ of the generator matrix follows from the combinatorial structure of K_4 alone.*

3.5 The proton–electron mass ratio μ

The PDL prediction for the proton–electron mass ratio is $\mu \approx 1836.152\,670$, a parameter-free result consistent with the experimental value $1836.152\,673\,43(11)$ at the level of 1.8×10^{-9} [47–49]. This is the programme’s primary near-term falsifiable prediction; current precision measurements can test it at 0.001 ppm.

4 The Three Gates: Critical Proofs

4.1 Gate 1: Axiomatic derivation of 155/11017

Theorem 4.1 (Gate 1, [48]). *The amplitude 155/11017 appearing in the mass-ratio correction formula is derivable from PDL axioms C1–C4 alone, via the AB coupling criterion. Specifically: the stability criterion AB has been verified exhaustively over all 768 admissible configurations; in every stable case, $P_2/P_1 = -1$ exactly; and the combinatorial count yields 155/11017 without any free parameter.*

4.2 Gate 2: The QCD correction coefficient $a = 2$

Theorem 4.2 (Gate 2, [49]). *The QCD correction coefficient $a = 2$ in the mass-ratio formula $\mu = \frac{155}{11017}(1 - 2\kappa^2)\frac{m_{\text{iso}}}{m_p}$ is structurally forced by axiom C1 (minimal binary pulsation): the factor 2 counts the two distinct pulsation branches of the proton’s valence core. Furthermore, $m_{\text{iso}} = m_d - m_u$ is the unique irreducible external parameter at the PDL–QCD interface; it cannot be derived from C1–C4 alone.*

4.3 Gate 3: $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$

Gate 3 was first established in D31 under a linear bridge inversion hypothesis. D36 strengthened this proof by: proving that the independence of gravitational engagements ($\sigma_0 = 0$) follows exactly from AB, without any additional assumption; proving that $\text{Tr}(A) = 0$ from the combinatorial structure of K_4 alone; and reducing the remaining hypothesis to H1 only (H2 was absorbed into the proof). D39 further reduced H1 to H3: given the Indifference Lemma, H1 follows as a theorem. D42 then proved H3 from C1–C4 unconditionally. See Theorem 3.7 for the full statement.

5 Derived Dynamical Equations

5.1 The Schrödinger equation (D32)

Theorem 5.1 ([51]). *In the continuum limit of the PDL coherence-cycle dynamics, the AB coupling criterion forces the amplitude evolution to satisfy the Schrödinger equation $i\hbar \partial_t \psi = -\frac{\hbar^2}{2m_e} \nabla^2 \psi + V\psi$. The diffusion parameter $b = -i\hbar/(2m_e)\partial_x^2$ is uniquely forced by the AB stability requirement; no free parameter enters.*

5.2 The Dirac equation (D33)

Theorem 5.2 ([52]). *Among the eight candidates for the time-reversal operator T acting on the $\text{SL}(2, \mathbb{C})$ pulsation of K_4 , the unique admissible choice consistent with AB is $T = -i\sigma_2$ (where σ_2 is the second Pauli matrix). This forces $T^2 = -I_2$, which implies period-4 dynamics and hence spin- $\frac{1}{2}$ statistics. The Clifford algebra of Dirac matrices then follows as a theorem (residual $< 0.001\%$), and the non-relativistic limit reproduces the result of Theorem 5.1 exactly.*

5.3 Born’s rule (D34 and D46)

Theorem 5.3 (Level 1, [53]). *Let $\sigma \in \{1, -1\}$ be the combinatorial signature of a coherence cycle in the PDL framework. Under the apparatus symmetry assumption, the leakage probability satisfies $p_{\text{coh}} \propto \sigma^2$. The five Gleason axioms (countability, null-assignment, covariance, normalisation, and regularity) are verified for this measure. By the Gleason uniqueness theorem for spin- $\frac{1}{2}$ systems, the probability assignment is uniquely given by Born’s rule $P(\psi) = |\langle \psi | \phi \rangle|^2$.*

Theorem 5.4 (Level 2, OP4 resolved, [65]). *The $\text{U}(1)$ phase freedom $\psi \mapsto e^{i\alpha}\psi$ of quantum mechanics is a structural consequence of the K_4 pulsation, not an independent postulate. Specifically: (Proposition 1) the global sign equivalence $s \sim -s$ of K_4 coherent configurations, combined with $T^2 = -I_2$ (D33),*

generates a canonical $U(1)$ action on the amplitude space $\mathcal{H}_{\text{cyc}} \cong \mathbb{C}^2$, whose orbits are the fibres of the Hopf fibration $S^1 \hookrightarrow S^3 \rightarrow S^2$; (Proposition 2) the $(A) \wedge (B)$ stability criterion is $U(1)$ -invariant, verified algebraically over all 8 coherent configurations; (Proposition 3) the unique $U(1)$ -equivariant probability measure on S^3 consistent with the Level 1 Gleason axioms is Born's rule $P = |\langle \theta | \psi \rangle|^2$ on the base $\mathbb{P}^1(\mathbb{C})$. This completes the quantum dynamics layer of the PDL programme.

5.4 The Einstein coupling equation (D35 and D48)

Theorem 5.5 ([54, 67]). Under Gate 3 (now unconditional via D42), the PDL gravitational coupling satisfies $G_{\text{eff}}(N) = G_{\text{Newton}}$, and the PDL Einstein field equation takes the form $G_{\mu\nu}^{\text{eff}} = 8\pi G(N) C_{\text{coh}}$, where $C_{\text{coh}} = C^{(\text{mass})} + C^{(\text{spin})} + C^{(\text{orb})} + C^{(\text{leak})}$ is the coherence stress-energy tensor with explicit components derived in D48. The Hubble tension ratio $H_{0,\text{local}}/H_{0,\text{CMB}} = 1.085868$ versus the PDL prediction 1.085935 gives a 0.006% relative error with zero free parameters.

6 Black Hole Thermodynamics

6.1 The area law as an unconditional theorem (D37)

Theorem 6.1 (BH-1, [56]). The information-theoretic entropy of a PDL closure is carried entirely by its active surface R_{surf} , not by its total relational budget R_{tot} . Formally: (1) the valence sector has $S_{\text{val}} = k_B \ln 1 = 0$ [48]: the AB-stable valence configuration is unique; (2) the sea sector is excluded from stationary coupling [48], hence $S_{\text{sea}} = 0$; (3) the total entropy is $S = k_B \ln R_{\text{surf}}$. This is an unconditional theorem; it requires no external geometric assumption and no hypothesis beyond C1–C4 and D29.

Corollary 6.2 (PDL area law, [56]). Under Proposition 6.1 and the identification of R_{surf} with the physical horizon area via $G_{\text{eff}}(N)$ (Gate 3, Theorem 3.7), the Bekenstein–Hawking area law $S \propto A$ is a structural consequence of the four PDL axioms. It is not an independent postulate.

6.2 The Bekenstein–Hawking formula (D38)

Theorem 6.3 (BH-2, [57]). Under Gate 3 ($G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$, now unconditional), substituting the PDL effective gravitational coupling into the Bekenstein–Hawking formula $S_{\text{BH}} = k_B c^3 A / (4\hbar G(N))$ yields the algebraic identity $S_{\text{BH}}/k_B = 4\pi (M_{\text{eff}}/M_{\text{Pl}})^2$, where $M_{\text{Pl}} = (\hbar c / G_{\text{PDL}})^{1/2}$. This identity is verified numerically to relative error $< 10^{-15}$ over ten decades of black hole mass. The coefficient 4π is not a free parameter; it is fixed by the Schwarzschild geometry.

Three falsifiable, parameter-free predictions arise from Theorem 6.3 [57]: (1) CMB-epoch entropy suppression by a factor ≈ 0.848 at $N_{\text{CMB}} = 40$; (2) primordial black hole survival threshold shifted upward by 11.89% relative to GR, testable with Fermi-LAT via BLACKHAWK/ISATIS [64]; (3) redshift-dependent S/E ratio in AGN populations traceable with DESI/Euclid.

D45 v2 — complete Hawking spectrum. The original PBH threshold prediction of D45 has been strengthened by a full spectral calculation using GAMMAPBHLOTTER and BLACKHAWK: the predicted photon spectral peak shifts from 130.1 MeV (GR) to 117.5 MeV (PDL), a -9.65% shift, giving a second, independent spectral signature alongside the $+11.89\%$ mass-threshold shift, both testable against existing Fermi-LAT data.

7 Black Hole Physics II: Soft Hair, Behavioural Universality, and the Microstate-Counting Problem (D64–D65)

Documents D64 and D65 extend the black hole physics of Section 6 in a direction that is deliberately *not* a direct derivation of new numerical results, but a systematic, honestly delimited exploration of what

the axioms C1–C4 do and do not currently support at the scale of a macroscopic collapsed object. Both documents are written to be self-contained for a reader from the black hole thermodynamics or soft-hair literature who is not otherwise familiar with PDL.

7.1 The soft-hair correspondence and its negative result (D64)

Theorem 7.1 (Surface degeneracy, recalled from D50 [69]). *The active surface of a PDL proton closure carries $\Omega_{\text{surf}} = 4^{R_{\text{surf}}}$ statistically independent microstates at fixed global quantum numbers (the proton quintuplet), while the interior carries none ($\Omega_{\text{val}} = 1$, Theorem 6.1).*

Conjecture 7.2 (Soft-hair structural correspondence, [11]). The logical architecture of Theorem 7.1 — a macroscopic object fully characterised by a small number of global invariants, yet carrying a combinatorially large space of boundary degrees of freedom invisible to any measurement of those invariants alone — is structurally analogous to the soft-hair programme of Hawking, Perry and Strominger [79, 80], in which black holes carry an infinite-dimensional space of supertranslation and superrotation charges at fixed mass, charge and angular momentum. This is presented explicitly as a structural analogy with independent motivation on both sides, following the cross-framework identification protocol of the PDL–OFN bridge [7, 73], and *not* as a claim of mathematical identity.

Proposition 7.3 (Mass-independent negative result, [11]). *A naive extrapolation of the PDL per-relation degeneracy to a macroscopic black hole, via combinatorial pairing of the N engaged nucleons, fails the Bekenstein–Hawking entropy value by 36.7 orders of magnitude. This discrepancy is proved to be mass-independent by algebraic construction (both quantities scale as N^2 with fixed prefactors) and is verified numerically to three significant figures for four black-hole masses spanning eight orders of magnitude: $1 M_{\odot}$, M87* ($6.5 \times 10^9 M_{\odot}$), Sagittarius A* ($4.297 \times 10^6 M_{\odot}$), and the GW150914 remnant ($62 M_{\odot}$) — the ratio is 4.68, 4.67, 4.68, and 4.67×10^{36} respectively.*

A surface-restricted count (counting only nucleons on a two-dimensional active surface, $N_{\text{surf}} \sim N^{2/3}$, rather than the full volume) narrows the solar-mass discrepancy to 1.4 orders of magnitude, but this improvement is shown in D64 *not* to generalise: the identical construction applied to M87* leaves a discrepancy of 7.9 orders of magnitude, nearly six times worse than the solar-mass case. This sharpens rather than weakens the conclusion that the relevant microscopic degrees of freedom are areal in some deeper sense not captured by nucleon counting at any fixed geometric scale.

7.2 Behavioural universality between the proton and the black hole horizon (D65)

D65 is a direct sequel to D64. Rather than attempting to close the 36.7-order-of-magnitude gap of Proposition 7.3, it asks a prior question: independently of any absolute microstate count, does PDL predict shared *behaviours* between its two most extreme closures — the isolated proton and the macroscopic black hole horizon — that are not visible in either limit alone?

Theorem 7.4 (Two forced coherence defects, [12]). *Any closed spherical PDL surface, as an unconditional consequence of the $\text{SL}(2, \mathbb{C})$ pulsation structure of K_4 (D33) rather than of any continuum approximation, requires an internal state space $\mathcal{H}_{\text{cyc}}$ richer than a binary sign. This forces exactly two coherence defects on the closed surface, never zero, by a combinatorial analogue of the Poincaré–Hopf theorem. The result is verified independently by three distinct methods: spinor relaxation, exact discrete Gauss–Bonnet computation, and direct holonomy calculation in arbitrary-precision arithmetic. This gives a structural, axiom-level account of why no black hole has ever been observed with exactly zero spin, complementing — without duplicating — the Kerr/CFT correspondence of Guica, Hartman, Song and Strominger [81], whose near-horizon $\text{AdS}_2 \times S^2$ geometry shares the same S^2 that already governs the topological origin of the exponent 18 at the proton scale (D23 v2).*

Theorem 7.5 (Curvature-migration exponent, [12]). *The polar curvature deficit of a PDL closure migrates towards the equator as the spin parameter χ increases towards extremality, with asymptotic exponent*

exactly 1 (not 2), once floating-point noise is eliminated by arbitrary-precision arithmetic. This is a genuine, openly reported tension with the third law of black hole thermodynamics, rather than a resolved result (Open Problem 16.37).

Conjecture 7.6 (Nucleon-dissolution mass threshold, [12]). Combining the previously established nucleon saturation number $Z_{\text{sat}} = 20$ (D22) with the standard nuclear force range gives a critical mass $M^* \approx 4.3 M_\odot$, at which the density required for nucleon dissolution coincides with the Schwarzschild condition. This value falls inside the observed compact-object mass gap without having been tuned to do so; a sensitivity scan over the nuclear force range $r_c \in [1.0, 3.0]$ fm shows M^* spans $[2.5, 5.0] M_\odot$ for $r_c \in [1.39, 2.21]$ fm, overlapping the pion Compton wavelength $\hbar/(m_\pi c) \approx 1.4$ fm.

Remark 7.7 (Seven documented negative results, [12]). D65 reports seven independent, precisely diagnosed failures of candidate mechanisms for an absolute microstate count or a Bardeen–Thorne-type spin-limiting mechanism: rigid-body inertia (kinematically impossible, requires $\beta = 2 > 1$); two-horizon thermodynamics (inconsistent with the first law of inner-horizon thermodynamics); geometric curvature saturation (remains below 11% of its absolute bound even at $\chi_f = 0.9999$); spin stress-energy suppression (non-zero but far too small and too gently varying, D48v3); a spurious coincidence with the 126 nuclear magic number (a category error, comparing an entity count with a relational quantity); polytropic pressure support with $\Gamma = 5/3$ (admits no maximum mass at all, unlike the $\Gamma = 4/3$ relativistic case); and a spin-horizon-scale correlation that does not survive extension from four to seven well-measured events. Three of these seven trace to a single common cause, formalised as Open Problem 16.38.

Conjecture 7.8 (Behaviour-versus-magnitude principle, [12]). At the black hole scale, every PDL result that has survived independent verification concerns a *behaviour*: what configuration is topologically permitted or forbidden (Theorem 7.4), what functional form a law must take near a boundary (Theorem 7.5), what threshold a phase transition must respect (Conjecture 7.6), or how a known quantity scales with a control parameter. Every attempt to instead predict an absolute *magnitude* — an entropy value, a coupling strength, a maximum supportable mass from first principles — has failed, traceable in every case examined either to a hidden circularity in importing G through a length or mass conversion, or to the unresolved extension of homogeneous-regime theorems to the inhomogeneous, symmetry-broken regime a real horizon occupies. This is proposed as a structural property of the current theory, delimiting its domain of validity at the black hole scale, not as a temporary shortage of successful calculations.

7.3 The falsifiable reference target: GW250114

The observational reference target adopted for the black hole physics II programme is GW250114 (SNR = 80): remnant mass $M_f = 62.7 \pm 1.0 M_\odot$, spin $\chi_f = 0.68 \pm 0.01$, ringdown frequencies $f_{220} = 247 \pm 6$ Hz and $f_{221} = 249^{+8}_{-9}$ Hz, and damping times $\gamma_{220} = 221^{+39}_{-32}$ Hz and $\gamma_{221} = 708^{+116}_{-107}$ Hz. Using the combinatorial-to-physical time anchor GM_f/c^3 alone, the PDL ringdown timescale reproduces the correct order of magnitude (≈ 0.2 ms) without fitting, consistent with the behaviour-versus-magnitude principle: the scaling is captured, the absolute normalisation is not yet derived.

8 The Coherence Stress-Energy Tensor (D48)

8.1 Double determination of the coherence volume

Theorem 8.1 (Double determination of V_C , [67]). *The coherence volume $V_C = (4/3)\pi[\hbar/(m_p c)]^3$ is an unconditional theorem of C1–C4. The proof proceeds via Chain 1 (D10a): the pulsation frequency $\omega_p = m_p c^2/\hbar$ forces $\lambda_C = \hbar/(m_p c)$ and $V_C = (4/3)\pi\lambda_C^3$, with $n = 3$ from D23. The geometric argument (Chain 2 of D48v1/v2) is a consistency check, not an independent proof: the relational unit invoked in Chain 2 is not derived from the axioms independently of Chain 1 (D48v3, Remark 3.1).*

8.2 The coherence mass density

Theorem 8.2 ([67]). *The coherence mass density of an ensemble of N nucleons is*

$$\rho_{\text{coh}}(N) = \sigma(N) \cdot \frac{R_{\text{surf}} \cdot m_p}{V_C \cdot R_{\text{tot}}}, \quad (2)$$

where $\sigma(N) = 1 - (1 - \kappa)^N \in \mathbb{Q}(\sqrt{5})$. This is an unconditional theorem of C1–C4.

8.3 Components of the coherence stress-energy tensor

Theorem 8.3 (Explicit form of C_{coh} , [67]). *The coherence stress-energy tensor appearing in the PDL Einstein equation decomposes as $C_{\text{coh}} = C^{(\text{mass})} + C^{(\text{spin})} + C^{(\text{orb})} + C^{(\text{leak})}$, with the following status for each component:*

1. $C_{\mu\nu}^{(\text{mass})} = \rho_{\text{coh}}(N) c^2 u_\mu u_\nu$ (*unconditional theorem*).
2. $C_{\mu\nu}^{(\text{spin})} = 0$ in the homogeneous ground state (*theorem*, D48v3; proved by $\langle S^i \rangle = 0$ and $\langle S^i S^j \rangle = \frac{1}{4} \delta^{ij}$ over all 8 coherent configurations of K_4 ; Weyssenhoff–Raabe traceless part vanishes exactly; OP-spin resolved).
3. $C_{\mu\nu}^{(\text{orb})} = m_p \langle j_\mu j_\nu \rangle / \rho_{\text{coh}}$ (*theorem*, state-dependent via D32; connection to London equation via D46 conjectured as OP-London).
4. $C_{\mu\nu}^{(\text{leak})} = -C \eta_L^{18} (m_p c / \hbar)^2 g_{\mu\nu}$ (η_L^{18} is a *theorem* from D17 and D30; the prefactor C is *open* as OPI-D35, bounded observationally at $C \approx 8.16 \times 10^{-46}$ from Planck 2020).

Corollary 8.4 (PDL Einstein equation with explicit source, [67]). *The PDL Einstein field equation with complete explicit source is:*

$$G_{\mu\nu}^{\text{eff}} = 8\pi G(N) \left[\rho_{\text{coh}} c^2 u_\mu u_\nu + C_{\mu\nu}^{(\text{spin})} + C_{\mu\nu}^{(\text{orb})} - C \eta_L^{18} (m_p c / \hbar)^2 g_{\mu\nu} \right]. \quad (3)$$

The constant C is bounded observationally but remains to be derived from C1–C4 (OPI-D35).

9 Equation of State of the Coherence Fluid (D54)

Resolved Problem 1 (OP-pressure, resolved by D54 [1]). *The equation of state $w = P/(\rho_{\text{coh}} c^2)$ of the PDL coherence fluid is an unconditional theorem of C1–C4.*

Theorem 9.1 (Complete equation of state, [1]). *In the homogeneous ground state, the equation of state of the PDL coherence fluid is*

$$w(N) = -\frac{\rho_\Lambda}{\rho_{\text{coh}}(N)}, \quad (4)$$

where $\rho_\Lambda = \Lambda_{\text{PDL}} c^2 / (8\pi G_{\text{PDL}}) = 5.835 \times 10^{-27} \text{ kg m}^{-3}$ and $\rho_{\text{coh}}(N)$ is the coherence density (Theorem 8.2). Every quantity in (4) is an unconditional theorem of C1–C4; there is no free parameter.

The four components of the equation of state are:

1. $w_{\text{mass}} = 0$ exactly: pressureless dust (D48v3 Thm 4.1, unconditional theorem).
2. $w_{\text{spin}} = 0$ exactly: isotropic spin distribution over K_4 (D48v3 Thm 5.3, OP-spin resolved, unconditional theorem).
3. $w_{\text{orb}} = 0$ exactly: London gauge forced by C4 (D49 Thm 1, unconditional theorem).
4. $w_{\text{leak}} = -1$ intrinsically: cosmological constant structure (D48v3 Prop. 8.1, unconditional theorem).

Corollary 9.2 (Two physical regimes, [1]). *(a) Coherence and nuclear densities* ($N = 1\text{--}200$, $\rho_{\text{coh}} \sim 10^{17}\text{--}10^{18} \text{ kg m}^{-3}$): $|w(N)| \sim 10^{-44}$. *The PDL coherence fluid behaves as pressureless cold matter, identical in equation of state to cold dark matter. (b) Cosmological scales:* $\rho_{\Lambda}/\rho_{\text{crit}} = 0.6838$, in agreement with $\Omega_{\Lambda}^{\text{obs}} = 0.685$ (Planck 2020 [82]) to 0.17%, with no cosmological parameter as input.

The PDL coherence fluid thus provides a unified description of two phenomena that, in the Λ CDM concordance model, require two distinct and unrelated postulates (cold dark matter and cosmological constant). Both emerge from the same combinatorial fluid sourced by the proton quintuplet under axioms C1–C4, distinguished only by the density regime.

10 Nuclear Stability and the Periodic Table

10.1 The neutron survival condition (D40)

Theorem 10.1 (Regime I, [59]). *The identity $T \equiv (\Delta n + 1)^2 = 1.010414 \approx 1$ is a consequence of the proton quintuplet. The neutron survival condition $N_{\min}(T, Z) = \lceil n_1/2 \rceil$ therefore entails $N_{\min}(Z) = Z$ for all $Z \leq 20$, exactly. This is an unconditional theorem; no hypothesis beyond the quintuplet values is required.*

10.2 Conflict saturation (D40)

Theorem 10.2 (Saturation, [59]). *For all $Z > 20$, the total proton–proton conflict saturates: $C(Z > 20) = 190 T_{pp}$ exactly. Every nucleus with $Z > 20$ presents the same total conflict to its neutron sea, independently of the number of protons. The observed increase of $N(Z)$ beyond $Z = 20$ is driven not by increasing conflict but by the structural requirement that each neutron maintains its own coherence.*

10.3 Spin–orbit splitting from axioms (D47)

Theorem 10.3 (OP13 resolved, [66]). *The value $\Delta n = 4$ is the unique element of $\{0, 4, 8, 12, \dots, 32\}$ such that the discriminant of the quasi-completeness equation $3n_u^2 + (2\Delta n - 3)n_u + \Delta n(\Delta n - 1) - 1860 = 0$ is a perfect square. The discriminant equals $149^2 = 22201$ for $\Delta n = 4$ and fails to be a perfect square for all other admissible values (exhaustive verification). This forces $n_u = 24$, $n_d = 28$, and $s = \Delta n/(2n_u) = 1/12$ as unconditional theorems of C1–C4.*

Lemma 10.4 (Mirror Lemma, [66]). *The harmonic oscillator PDL levels of the proton and neutron are isomorphic: $n_u(n) = n_u(p) = 24$ and $n_d(n) = n_d(p) = 28$. Proton and neutron share the same splitting $1/12$, the same level order, and the same degeneracies. Verified analytically for the first 20 levels; zero exception.*

10.4 Sub-shell filling rates and the periodic table (D47)

Theorem 10.5 (OP14 resolved, [66]). *The sub-shell filling rates for all $Z = 1\text{--}82$ are:*

$$r_{\text{exc}}(Z) = \begin{cases} 0 & \text{if } Z \in \{28, 50, 82\} \text{ (magic closures, from OP13),} \\ 1 & \text{otherwise (mirror filling, ratio } 2/2 = 1 \text{ independent of degeneracy } d). \end{cases} \quad (5)$$

Verified exhaustively for 31 sub-shell boundaries (31/31), zero free parameter, zero external input.

Corollary 10.6 (Valley of stability as an unconditional theorem, [66]). *The formula $N_{\min}(Z) = Z$ for $Z \leq 20$ and $N_{\min}(Z) = \lfloor 20 + r_{\text{exc}}(Z) \cdot (Z - 20) \rfloor$ for $Z > 20$ is an unconditional theorem of C1–C4, with $r_{\text{exc}}(Z)$ given by Theorem 10.5. The periodic table (valley of stability for $Z = 1\text{--}82$) is a corollary of the four PDL axioms.*

10.5 Nuclear spectroscopy confrontation (D41) and the closure multiplicity theorem (D56)

Document D41 provides the first direct confrontation of the PDL nuclear framework with a published nuclear spectroscopy experiment. The experiment of Ha et al. [83] measures the first-excited-state lifetimes in ^{84}Mo and ^{86}Mo . Three independent observables are consistent with the PDL structural identity $T = (\Delta n + 1)^2 = 25$: (1) the BE2 ratio (22% discrepancy, within the 33% experimental uncertainty on ^{84}Mo); (2) the wave-function component ratio $104/53 = 1.962$ within 2% of the PDL prediction 2.000; (3) the Shannon entropy difference (25%). Two falsifiable predictions for $^{88}\text{Ru}/^{90}\text{Ru}$ and $^{92}\text{Pd}/^{94}\text{Pd}$ follow from Conjecture H_B (OP17). The precision lifetime measurements of Escudeiro, Recchia, Lenzi et al. [84] in the $f_{7/2}$ mirror nuclei ^{47}Cr – ^{47}V and ^{49}Mn – ^{49}Cr provide additional structural motivation.

Open Problem OP-D41-1-A has been fully resolved by D56 [3]:

Theorem 10.7 ($N_{\text{comp}}(k) = k$, OP-D41-1-A resolved, [3]). *Let a k p- k h nucleus at the boundary of an island of inversion be modelled as k independent $(A) \wedge (B)$ -coupled proton–neutron interface units (Definition in D56). The number of maximal C1+C2+C3+C4-admissible partial closures satisfies:*

$$N_{\text{comp}}(k) = k. \quad (6)$$

This is an unconditional theorem of C1–C4, proved via three lemmas: L1 (uniqueness of the maximal closure on one unit, D16a [31]); L2 (disjointness of the mixed-triangle sets of k distinct $(A) \wedge (B)$ couplings, D29 [48]); L3 (C3-irreducibility of the k -unit assembly, axiom C3 directly). Verified numerically for $k = 1, \dots, 5$. Corollary: $R_{\text{surf}}(k) = kT$ for a k p- k h nucleus.

Conjecture 10.8 (H_B — Closure multiplicity, [60]). For a nucleus at the boundary of an island of inversion dominated by a k p- k h intruder configuration, $B(E2) \propto N_{\text{comp}}(k) = k$ (Theorem 10.7), yielding the ratio:

$$\frac{B(E2)[k_1\text{p-}k_1\text{h}]}{B(E2)[k_2\text{p-}k_2\text{h}]} = R_{\text{PDL}} \cdot \frac{k_1}{k_2}, \quad R_{\text{PDL}} = \frac{2T}{(\Delta n + 1)^2} = 2.021. \quad (7)$$

For $k_1 = 8$ (^{84}Mo) and $k_2 = 4$ (^{86}Mo): $R_{\text{PDL}} = 2.021$, within 0.58σ of the experimental ratio 2.461 ± 0.756 [83]. Note: $N_{\text{comp}}(k) = k$ is now a theorem; the remaining gap is the formal identification of the $E2$ operator in the PDL formalism (Open Problem OP-E2-PDL).

11 Cosmology

11.1 The density function $\sigma(N)(N)$

For an ensemble of N protons (or nucleons), the effective coherence engagement fraction is $\sigma(N) = 1 - (1 - \kappa)^N \in \mathbb{Q}(\sqrt{5})$, where κ is the surface-to-total ratio of the proton quintuplet [43, 45]. As $N \rightarrow \infty$, $\sigma(N) \rightarrow 1$, recovering κ_G in the dense limit. As $N \rightarrow 0$, $\sigma(N) \approx N\kappa$, giving a suppressed coupling at low densities.

11.2 The Hubble tension resolution

Theorem 11.1 ([46, 54]). *Let $n_6 = 6n_-/n_+$, where n_- is the neutron magnetic moment in nuclear magnetons and $n_+ = 10992$ is the total relational content of the neutron. Then $n_6 \approx 40.102$, so $N_{\text{CMB}} = \lfloor n_6 \rfloor = 40$. The CMB-epoch Hubble constant is $H_{0,\text{CMB}} = 67.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$, which agrees with the Planck 2018 measurement to within 0.27σ . The ratio $H_{0,\text{local}}/H_{0,\text{CMB}} = \sigma(N)(N_{\text{local}})/\sigma(N)(N_{\text{CMB}})$ reproduces the observed Hubble tension at 0.006% with zero free parameters.*

12 The PDL–QCD Interface

The unique irreducible external parameter of the programme is $m_{\text{iso}} = m_d - m_u$, the up–down quark mass difference. This quantity arises at the PDL–QCD boundary (Theorem 4.2) and cannot be derived from the relational axioms alone.

Current value and uncertainty. The PDL structural derivation yields $m_{\text{iso}} \approx 2.532$ MeV, to be compared with the FLAG lattice QCD result $m_{\text{iso}} = 2.529 \pm 0.030$ MeV [49]. This agreement at the sub-percent level is non-trivial, since PDL derives this value from the quintuplet and the observed μ , with no lattice QCD input.

Residual. A 47 ppm residual remains in the mass-ratio formula; the structural factor g accounting for it has not yet been identified (Open Problem 16.8).

13 Causal Chain Closure and Geometric Leakage (D43–D44)

Documents D43 [62] and D44 [63] together close the causal chain from axioms to Newton’s constant.

D43 — Geometric leakage and OP-A resolution. D43 establishes the computational backbone of the programme (Python/SymPy at 50-decimal precision) and proves the geometric leakage parameter as an unconditional theorem. The general analytic formula for the boundary energy is:

$$E_{\text{bord}} = A \cdot n_{u\text{-cores}} + B \cdot n_{d\text{-cores}} + (\Delta n + 1)^2, \quad A = 55, \quad B = 194, \quad (8)$$

where $c = 3$ is forced by the K_4 3-regularity. This yields $\varepsilon_{\text{geom}}(p) = 329/10087$ and $\varepsilon_{\text{geom}}(n) = 468/9960$ as unconditional theorems of C1–C4 (OP-A resolved).

D44 — Filter factor k and OP-B resolution. D44 derives $k = 0.921716$ from C1–C4 via the trace structure of K_4 and the S_4 -equivariance constraints proved in D42. The complete causal chain is:

$$\text{C1–C4} \rightarrow K_4 \rightarrow \text{quintuplet} \rightarrow \varepsilon_{\text{geom}} \rightarrow \varepsilon_G = \varepsilon_{\text{geom}} \times k^{18} \rightarrow G.$$

No remaining open problem exists in this chain. As a corollary, OP8 is resolved: the 17 ppm residual of D28 is attributed to the CODATA measurement precision of G , not to any structural gap.

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Note N01 [7] established that $\beta_1(K_4) = 3$ (PDL) and $b_1(\Omega_{21}) = 3$ (OFN, under Hamming distance-1 adjacency) are the same topological invariant viewed from two independent frameworks, and identified Open Problem OP-OFN-1 (a formal link between the three PDL leakage cycles and the three OFN fermion generations). An in-depth exchange with Oleg Evdokimov, held as Session 67 of the programme, produced five further precise connections, reported here for the first time and destined for Note N02, *From $\mathbb{Z}_3$ to Three Generations: A PDL–OFN Bridge* [73], currently in preparation.

- (i) **A mathematical identity, not a candidate analogy.** The tripartition $\{A, B, C\}$ of the complete tripartite graph $K_{2,2,2}$ — which is isomorphic to the line graph $L(K_4)$ — is exactly the set of the three perfect matchings of K_4 , itself an unconditional theorem of C1–C4 (D58 Lemma L2, D61). This is the first entry in the PDL–OFN comparison table with status *mathematical identity* rather than *candidate structural analogy*.

- (ii) **An exact algebraic identity in $\mathbb{Q}(\sqrt{5})$.** The relation $\varphi + \gamma/2 = 2$ holds exactly, where $\varphi = (1 + \sqrt{5})/2$ is the PDL golden ratio (appearing in $\kappa = 310\varphi/11017$) and $\gamma = 3 - \sqrt{5}$ is the spectral gap of Ω_{21} in OFN. The filter factor $k = [R\varphi]^{1/18}$ (with $R = 9758 \cdot 930 \cdot 310/(10087 \cdot 11017^2)$ rational) consequently lies in $\mathbb{Q}(\sqrt{5})^{1/18}$, a strictly larger field than $\mathbb{Q}(\sqrt{5})$ itself: γ enters k structurally through this identity, but k is not itself an element of $\mathbb{Q}(\sqrt{5})$.
- (iii) **A corrected automorphism computation.** $L(K_4) \cong K_{2,2,2}$ is verified computationally. An initial application of Whitney’s theorem ($\text{Aut}(L(G)) \cong \text{Aut}(G)$) to conclude $\text{Aut}(K_{2,2,2}) = S_4$ was incorrect: Whitney’s isomorphism theorem is known to admit exactly two exceptional cases, K_3 and K_4 , to which it does not apply. The correct result is $\text{Aut}(K_{2,2,2}) = S_2 \wr S_3$, of order 48 — the octahedral group O_h — capturing the internal S_3^2 action within each part of the tripartition, in addition to the external S_3 permutation of the three parts already established in D58/D59.
- (iv) **A dimensional echo.** K_4 has 4 vertices (entities) and 6 edges (relations), $4 + 6 = 10$ elements in total, matching $\dim \mathcal{P}(1, 3) = 4$ (translations) + 6 (Lorentz generators) = 10, a correspondence already noted independently in D35 and D61.
- (v) **A reinforced verification of the hadronic–cosmological identity.** The script `PDL_N02_identity_lockdown_v2` (not yet deposited, to be included with N02) verifies the identity $n_u - 1 = p_{k_1} = 23$ (D47+D51) with 13/13 checks passed and 0 failures, at an isolation level of 0.265% against 378 pre-registered candidate integer pairs, and against a null model at 0.059%.

Several proposals made during the exchange were evaluated against the same standard as PDL’s own results and found not to meet it; these negative outcomes are documented with the same rigour as the five positive connections above, following the epistemic conventions of Section 1.3. In particular: a candidate formula $\varepsilon_{\text{geom}} \approx 2k\gamma/43$ is retained only as a numerical curiosity (the subtraction $2^6 - |\Omega_{21}| = 43$ has no independent PDL justification); a candidate “GM-scale” relation between muon and tau mass ratios was withdrawn by Oleg Evdokimov himself upon discovering an algebraic redundancy ($(\sqrt{2})^2 = 2$ reduces three apparently independent parameters to two); and a numerical coincidence involving 1682/11017 was likewise self-identified and withdrawn. Negative results volunteered by a collaborator on their own proposals are recorded here as scientific contributions in exactly the same spirit as the negative results reported internally throughout the PDL corpus (D64, D65, Section 7.2).

Two manuscripts co-authored with Oleg Evdokimov, Bachani and Ryss — *Three Roads to the Periodic Table* and *unified_theory.pdf* — were reviewed against the PDL epistemic standard during Session 67; both currently present Hypothesis $H_{\text{SU}2}$ as a conjecture, whereas D60 establishes it as an unconditional theorem of C1 and C2 alone, a correction communicated to the co-authors and pending revision.

15 Epistemic Architecture

Table 2 Nine-layer epistemic architecture of the PDL programme (updated for v20). Each layer lists the status of its key claims and the primary documents supporting them.

Layer	Domain	Key claims and status	Primary documents
L0	Computational verification	Infrastructure: 18 filters at 50-decimal precision (D43). $\varepsilon_{\text{geom}}(p) = 329/10087$ and $\varepsilon_{\text{geom}}(n) = 468/9960$ proved (OP-A). $k = 0.921716$ proved (OP-B, D44). ε_G^B closed-form in $\mathbb{Q}(\sqrt{5})$ exact (OP8 resolved).	D43, D44

Layer	Domain	Key claims and status	Primary documents
L1	Axioms / combinatorial results	Theorems: K_4 unique [31]; $\nu_u = 24$ unique (D47 OP13); $18 = 6 + 5 + 4 + 3$ [42]; $\sigma = 930$ unique; $a = 2$ [49]; $\sigma(N) = 1 - (1 - \kappa)^N$ [43]; AB criterion [48]; $T = -i\sigma_2$ unique [52]; Clifford algebra [52]; period-4 [52]; $\sigma_0 \equiv 0$ [55]; $\text{Tr}(A) = 0$ [55]; $S_{\text{val}} = 1$ [56]; $\Delta n = 4$ unique (D47). No free parameter; no external input.	D01, D16a, D23, D29, D30, D33, D36, D37, D47
L2	QCD interface	Minimal, forced: $m_{\text{iso}} = m_d - m_u$ is the unique irreducible external input, forced by C1, not chosen. Conjecture: $m_{\text{iso}} \approx 2.532$ MeV.	D28, D30
L3	Cosmological coupling	Theorem: $\kappa = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$ unconditional (D42). $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ un- conditional.	D24, D31, D36, D39, D42
L4	Quantum dynamics [COMPLETE]	Theorems: Schrödinger [51]; Dirac [52]; Born Level 1 [53]; Born Level 2 + $U(1)$ (OP4 re- solved) [65]; London equation (OP- London resolved) [68].	D32, D33, D34, D46, D49
L5	Gravitational and black hole dynamics [COMPLETE]	Theorem (unconditional): $G_{\mu\nu}^{\text{eff}} = 8\pi G(N)C_{\text{coh}}$. The- orem: area law $S \propto R_{\text{surf}}$ unconditional [56]. Theorem: $S_{\text{BH}}/k_B = 4\pi(M_{\text{eff}}/M_{\text{Pl}})^2$ [57]. Theorem: coefficient $\frac{1}{4}$ (OP12 re- solved) [69]. C_{coh} explicit: all four components unconditional theo- rems [67, 70, 71] (D48v3, OP-spin resolved; dimensionally consis- tent).	D35, D36, D37, D38, D48, D49, D50
L5b	Equation of state [COMPLETE, D54]	Theorem OP-pressure resolved [1]: $w(N) = -\rho_\Lambda/\rho_{\text{coh}}(N)$; $w_{\text{mass}} =$ $w_{\text{spin}} = w_{\text{orb}} = 0$ exactly; $w_{\text{leak}} =$ -1 . $\Omega_\Lambda^{\text{PDL}} = 0.6838$ vs 0.685 (Planck 2020): 0.17% , no free pa- rameter. Two density regimes: cold matter ($ w \sim 10^{-44}$) and dark energy ($w \approx -0.62$).	D48, D49, D51, D52, D54

Layer	Domain	Key claims and status	Primary documents
L6	Nuclear stability / periodic table / closure multiplicity [COMPLETE + D56]	Theorems: $N_{\min}(Z \leq 20) = Z$ exact; $C(Z > 20) = 190 T_{pp}$ [59]; $\Delta n = 4$ unique (OP13) [66]; Mirror Lemma [66]; $r_{\text{exc}}(Z)$ 31/31 (OP14) [66]; $N_{\min}(Z)$ $Z = 1$ –82 unconditional corollary [66]; $N_{\text{comp}}(k) = k$ (OP-D41-1-A resolved) [3]. Conjecture H_B : $B(E2) \propto k$; open: OP-E2-PDL.	D22, D23, D40, D41, D47, D56
L7	Falsifiable predictions	$\mu \approx 1836.152\,670$ testable at 0.001 ppm; $m_{\text{iso}} \approx 2.532$ MeV vs. FLAG; PBH threshold $M_{\text{PDL}}^* = 5.706 \times 10^{14}$ g (+11.89%, D45); spectral cutoff at 93–104 MeV in IGRB (D45); $H_{0,\text{CMB}} = 67.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$; AGN $S/E \propto \sigma(N(z))$; $B(E2)$ ratio and N_{comp} ratio in $^{88}\text{Ru}/^{92}\text{Pd}$; $\lambda_{L,\text{PDL}}(N = 40) \approx 7.25 \times 10^{-15}$ m (D49).	D28–D30, D35, D38, D41, D45, D49
L8	Cosmological leakage [COMPLETE]	Theorems: $\beta_1(K_4) = 3$ (topology, D23); C1+C3 forces prime exponents (D51); indices $k_1 = 9$, $k_2 = 19$, $k_3 = 168$ exact (D51); bases $(1 - \kappa)$, $R_{\text{val}}/R_{\text{tot}}$, $(1 - \eta_L)$ uniquely identified (D52). $C = (1 - \kappa)^{997} \times (930/11017)^{23} \times (1 - \eta_L)^{67}$ at 0.17 ppm; $\Lambda_{\text{PDL}} \approx 1.089 \times 10^{-52} \text{ m}^{-2}$. Full chain C1–C4 $\rightarrow \Lambda_{\text{PDL}}$ complete. Independent numerical verification 0.41 ppm (D53).	D17, D23, D29, D30, D42, D51, D52, D53
L-DL	Life and consciousness (PDL-V, Part III)	Theorems (DL01): $S(K_4) \cong K_4$ (8/8); rarity law $f(n) = 2^{n+1-\binom{n}{2}}$ exact; $\delta^* = 1/2$ invariant; R_{weak}^1 forced for $n \geq 4$. Theorems (DL02): K_4 unique fixed point of S ; $n_{\text{life}}^* < n_{\text{consciousness}}^*$ (separation); $n_{\text{max}}^* < \infty$ (finiteness). Hierarchy $4 < n_{\text{life}}^* < n_{\text{consciousness}}^* < n_{\text{max}}^* < \infty$; unconditional theorem. Numerical values open (DL-OP1, DL-OP2).	DL01, DL02

Layer Domain	Key claims and status	Primary documents
L15 Gauge sector [COMPLETE — D46 + D57 + D58 + D59 + D60 + D61]	Theorems: $U(1)$ (D46, Hopf), $\sin^2 \theta_W^{(\text{tree})} = 1/4$ (D57), $\theta_W = 19\pi/119$ (D55), $G_{\text{eff}} = S_4/V_4 \cong S_3$ (D60), $SU(3)$ algebraic (D58+CKW), $SU(3)_c$ in 3 (D59), $SU(3) \times SU(2) \times U(1)$ theorem (D46+D60+D58+D59), $\text{Coh}(K_4) \cong \mathbb{Z}_2^3$ admissible transports (D61- α), $V_4 \hookrightarrow \text{Aut}(\mathbb{Z}_2^3)$ faithful (D61- β), $D_\mu = \partial_\mu - igA_\mu$ unique (D61- δ +Utiyama). Gauge structure and gauge dynamics derived. OP10-c resolved by D62 (M_Z/M_W theorem). Open: three generations (OP-D61-2), autonomous G_{gauge} (OP-D61-3).	D33, D46, D55, D57, D58, D59, D60, D61
L15b Gauge boson masses [D62 — new in v29]	Theorems: $M_Z/M_W = 1/\cos(19\pi/119)$ (D55+D57+D62); 4 Higgs decompositions in $\mathbb{Z}_2^3$; neutrinos exit at level 2; formal definition of m_B . Conjectures: $v = 14161/54 \cdot m_p$ [676 ppm]; M_Z/m_p [3.67 σ]; $M_H/m_p = 133.611$ [1.49 σ]. Open: OP-D62-1 (correction on v), OP-D62-2 (λ_H), OP-D62-3 (uniqueness of M_Z fixed point), OP-D62-4 ($\Delta\alpha_{\text{had}}$), OP-D62-5 (5^2 factor).	D55, D57, D61, D62
L16 Quark mass spectrum [D63 — new in v29]	Exact identity $n_u - 1 = p_{k_1} = 23$ (D47+D51, unconditional). H_mass: $m_d/m_u = 2401/1104$; $m_u = 2.155 \text{ MeV}$; $m_d = 4.687 \text{ MeV}$ (no free parameter beyond Δm_{iso}). H_sea: m_s, m_c, m_b, m_t within 2.5% PDG via residual-fluidity weighting; non-hadronisation of top quark as structural corollary. Open: OP-D63-1 (formal independence C2,C3,C4 in $\mathcal{Q}(K_n)$), OP-D63-2 (formal derivation of $w(n)$), OP-D63-3 (structural origin of effective sizes).	D31, D47, D51, D63

Layer	Domain	Key claims and status	Primary documents
L17	Black hole physics II: soft hair, behavioural universality [D64 + D65 — new in v30]	Structural analogy: soft-hair correspondence (D64), explicitly not a mathematical identity. Proposition (negative, mass-independent): nucleon-counting extrapolation fails Bekenstein–Hawking by 36.7 orders of magnitude (D64). Theorems (D65): exactly two coherence defects on any closed spherical closure, never zero; curvature-migration exponent = 1 near extremality (open tension with third law). Conjecture: dissolution threshold $M^* \approx 4.3 M_\odot$ inside observed mass gap. Seven further negative results organised under the behaviour-versus-magnitude principle. Open: OP-D64-1 (macroscopic microstate count), OP-D64-2 (relational c , μ^*), OP-D65-1 to OP-D65-3.	D22, D33, D37, D38, D48, D50, D54, D64, D65
L-OFN	PDL–OFN collaboration [Session 67 — new in v30]	Mathematical identity: $K_{2,2,2} = L(K_4)$ tripartition = three perfect matchings of K_4 (D58, D61). Theorem: $\varphi + \gamma/2 = 2$ in $\mathbb{Q}(\sqrt{5})$; $k \in \mathbb{Q}(\sqrt{5})^{1/18}$. Corrected: $\text{Aut}(K_{2,2,2}) = S_2 \wr S_3$ (order 48), Whitney’s theorem excepted for K_4 . $4 + 6 = 10 = \dim \mathcal{P}(1, 3)$. Lockdown script for $n_u - 1 = p_{k_1} = 23$: 13/13 PASS. N02 in preparation. Open: OP-OFN-1 (three generations, partially addressed by D57).	D57, D58, D59, D61, N01, N02

16 Open Problems

The following list supersedes the open-problem list of v29. Problems are classified by priority and domain. OP1, OP4, OP8, OP11, OP12, OP13, OP14, OP-London, OP1-D35, OP-pressure, OP-D57-1, OP-D59-2, OP-SU3, OP-D58-1, OP-D58-3, and OP-D41-1-A have been fully resolved and are retained for historical reference only. OP-D59-1 is partially addressed by D63 (see below). OP-D61-1 is partially addressed by D62. New in v30: five open problems are introduced by the black hole physics II programme (OP-D64-1, OP-D64-2, OP-D65-1, OP-D65-2, OP-D65-3); OP-D64-1 (the macroscopic microstate-counting problem, entry point: the $K_4 \leftrightarrow K_4$ coupling rule) and OP-D65-3 (the inhomogeneous, symmetry-broken regime) are jointly identified as the highest-priority entry points for the continuation of the programme at the black hole scale. OP-OFN-1 remains open, with the mathematical identity of Section 14(i) now available as additional structural context.

Open Problem 16.1 (OP1 — Resolved: Indifference Lemma H3). **Fully resolved by D42.** The Indifference Lemma (H3) has been derived from axioms C1–C4 via three lemmas: D42-L1 (Equiparticipation), D42-

L2 (Cross-Triangle Blindness), D42-L3 (S_4 -Equivariance). As a corollary, $\kappa = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$ is an unconditional theorem. The entire causal chain from C1–C4 to the Bekenstein–Hawking entropy is now fully axiomatic.

Open Problem 16.2 (OP1-D35 — Resolved: cosmological leakage constant C). [Fully resolved by D51+D52](#). The constant C in $C_{\mu\nu}^{(\text{leak})} = -C \eta_L^{18} (m_p c / \hbar)^2 g_{\mu\nu}$ is $C = (1 - \kappa)^{997} \times (930/11017)^{23} \times (1 - \eta_L)^{67}$, derived from C1–C4 via $\beta_1(K_4) = 3$ (D23), prime cycle lengths (D51 Lemma 2), quintuplet indices (D51 Lemma 3), and identification of the three leakage bases (D52). Agreement with Λ_{obs} : 0.17 ppm. The causal chain C1–C4 $\rightarrow \Lambda_{\text{PDL}}$ is complete.

Open Problem 16.3 (OP2 — Global uniqueness of the proton quintuplet). Prove that (24, 28, 930, 10087, 11017) is the *global* (not merely local) minimum of the coherence leakage functional over all admissible integer tuples. The local uniqueness conjecture has been verified computationally for all tuples within a neighbourhood of radius 10^4 ; the global statement remains open. **Priority:** high. **Entry point:** D16, D16b.

Open Problem 16.4 (OP3 — Algebraic proof of 155/11017). Provide a closed-form algebraic derivation of 155/11017 from the combinatorial structure of K_4 , replacing the exhaustive enumeration over 768 configurations in D29. The exhaustive verification is complete and rigorous, but an analytic proof would yield structural insight. **Priority:** medium. **Entry point:** D29.

Open Problem 16.5 (OP4 — Resolved: Born’s rule Level 2). [Fully resolved by D46](#). The U(1) phase freedom $\psi \mapsto e^{i\alpha}\psi$ has been derived from the K_4 pulsation structure via three propositions establishing the Hopf fibration structure, U(1)-invariance of the AB criterion, and uniqueness of the Born measure. The quantum dynamics layer is complete.

Open Problem 16.6 (OP5 — CMB angular power spectrum from PDL). Derive the CMB angular power spectrum C_ℓ from the PDL cosmological framework, using the $\sigma(N)$ -modified gravitational coupling. The scalar spectral index $n_s \approx 0.965$ and the tensor-to-scalar ratio r are the primary targets. **Priority:** medium. **Entry point:** D27, D35.

Open Problem 16.7 (OP6 — PDL prediction for gravitational wave polarisation). Determine whether the PDL Einstein equation (with the $\sigma(N)$ -modified coupling) predicts any deviation from GR in the polarisation or amplitude of gravitational waves at cosmological scales. **Priority:** medium. **Entry point:** D35, D38, D48.

Open Problem 16.8 (OP7 — Derivation of the 47 ppm residual in μ). Identify the combinatorial factor g responsible for the 47 ppm residual in the mass-ratio formula and derive it from C1–C4. Gate 1 (D29) and Gate 2 (D30) account for the leading structure. **Priority:** high. **Entry point:** D28, D29, D30, D43.

Open Problem 16.9 (OP8 — Resolved: closed-form for ε_G^B in $\mathbb{Q}(\sqrt{5})$). [Resolved by D44](#). The formula $\varepsilon_G^B = \kappa \cdot C / (6 + \kappa) \cdot (1 + 2/R_{\text{tot}}) \in \mathbb{Q}(\sqrt{5})$ is an exact closed-form expression. The 17 ppm residual relative to CODATA reflects the measurement precision of G , not a structural gap.

Open Problem 16.10 (OP9 — Extension to higher generations: muon and tau). Generalise the PDL proton architecture to the second and third generations of the Standard Model. The muon–electron and tau–electron mass ratios should be derivable from hierarchical extensions of the quintuplet. **Priority:** medium. **Entry point:** D01, D12.

Open Problem 16.11 (OP10 — Partially resolved: Weinberg angle (D55, D57); W/Z mass ratio open). [\$\theta_W\$ resolved by D55 \[2\]](#). The Weinberg angle is derived as an unconditional theorem of C1–C4: $\theta_W = 19\pi/119$, where $k_2 = 19$ (D51 theorem) and $N_{\text{total}} = 119$ (Lemma D, proved exhaustively). The predicted $\sin^2 \theta_W = 0.231196$ agrees with PDG 2024 at 0.48σ . [Tree-level foundation resolved by D57 \[6\]](#). The combinatorial tree-level value $\sin^2 \theta_W^{(\text{tree})} = \sin^2(\pi/R_e) = 1/4$ is an unconditional theorem of C1–C4: $R_e = 6$ is forced by D16a, and $1/4 = 192/768$ from D29. The exact value of D55 is the Berry-phase correction of this tree-level result: $\theta_W^{(\text{D55})} = \theta_W^{(\text{tree})} \cdot R_e k_2 / N_{\text{tot}} = (\pi/6) \cdot (114/119)$. The secondary prediction $\Delta m_{\text{iso}} = 2.446$ MeV (Prediction P10) is falsifiable by next-generation lattice QCD. Remaining open: (OP10-c) derive the W/Z mass ratio $M_W/M_Z = \cos(19\pi/119)$; (OP10-d)

experimental test of Δm_{iso} . **Note:** OP-D57-1 (formal proof of Lemma C1- V_4) has been resolved by D60; $H_{\text{SU}2}$ is now an unconditional theorem. **Priority:** medium. **Entry point:** D29, D33, D46, D51, D55, D57.

Open Problem 16.12 (OP11 — Resolved: dark energy as cosmological leakage). **Fully resolved by D51+D52 (synthesis D53).** The cosmological constant $\Lambda \approx 1.089 \times 10^{-52} \text{ m}^{-2}$ is derived from C1–C4 as $\Lambda_{\text{PDL}} = C \cdot \eta_L^{18} \cdot (m_p c / \hbar)^2$ where $C = (1 - \kappa)^{997} \times (930/11017)^{23} \times (1 - \eta_L)^{67}$, agreeing with the observational value to 0.17 ppm. The smallness of Λ is a combinatorial consequence of three prime-power suppressions imposed by C3, not a fine-tuning.

Open Problem 16.13 (OP12 — Resolved: Bekenstein–Hawking coefficient $\frac{1}{4}$). **Fully resolved by D50.** The coefficient $\frac{1}{4}$ in $S_{\text{BH}} = k_B A / (4\ell_P^2)$ is the stable fraction 4/16 per surface relation, proved from the $(A) \wedge (B)$ criterion (D29, 768/768) and the independence of surface couplings (H3, D42). $\Omega_{\text{surf}} = 4^{R_{\text{surf}}}$, $S_{\text{surf}} = k_B R_{\text{surf}} \ln 4 = 695.352 \text{ nats}$. Black hole thermodynamics layer complete.

Open Problem 16.14 (OP13 — Resolved: spin–orbit splitting from axioms). **Fully resolved by D47.** The value $\Delta n = 4$ is the unique element of $\{0, 4, 8, \dots, 32\}$ such that the quasi-completeness discriminant is a perfect square (149^2), forcing $n_u = 24$, $n_d = 28$, $s = 1/12$ as unconditional theorems of C1–C4.

Open Problem 16.15 (OP14 — Resolved: sub-shell filling rates $r_{\text{exc}}(Z)$ from axioms). **Fully resolved by D47.** The filling rates $r_{\text{exc}}(Z) = 0$ for $Z \in \{28, 50, 82\}$ and $r_{\text{exc}}(Z) = 1$ otherwise are derived analytically from C1–C4, verified exhaustively for 31/31 sub-shell boundaries. The valley of stability for $Z = 1\text{--}82$ is an unconditional corollary of C1–C4.

Open Problem 16.16 (OP15 — Elements $Z > 82$: superheavy nuclei and island of stability). Extend the valley-of-stability theorem (Corollary 10.6) to $Z > 82$. For $Z > 82$ the PDL prediction requires the full $\sigma(N)$ dependence, including relativistic corrections. Preliminary analysis suggests the PDL island of stability at $Z = 114\text{--}120$, $N = 172\text{--}184$. **Priority:** medium. **Entry point:** D40, D41, D47.

Open Problem 16.17 (OP-D41-1-A — Resolved: $N_{\text{comp}}(k) = k$). **Fully resolved by D56 [3].** The number of maximal partial closures on a $k\text{p-}k\text{h}$ nuclear interface surface is $N_{\text{comp}}(k) = k$ exactly (Theorem 10.7). Three lemmas suffice: L1 (uniqueness on one unit, D16a), L2 (disjointness of mixed-triangle sets, D29), L3 (C3-irreducibility of the assembly). Verified $k = 1, \dots, 5$.

Open Problem 16.18 (OP-OFN-1 — Formal link: 3 PDL leakage cycles $\leftrightarrow$ 3 OFN fermion generations). The joint PDL–OFN note N01 [7] established that $\beta_1(K_4) = 3$ (PDL) and $b_1(\Omega_{21}) = 3$ (OFN) are the same topological invariant from two independent frameworks. D57 [6] provides a partial resolution: the quotient $A_4/V_4 \cong \mathbb{Z}_3$ in $\text{Sym}(K_4) = S_4$ is the combinatorial origin of the invariant 3 in PDL. D59 strengthens this: the same triplet $(T_2) = \{k_1, k_2, k_3\}$ labels the three axes of the colour-charge carrier space W , tightening the structural link between the leakage cycles and the gauge sector. What remains open is the formal identification of this $\mathbb{Z}_3$ with the three fermion generations of the OFN framework. **Priority:** high. **Entry point:** N01, D51, D52, D57, D59.

Resolved Problem 2 (OP-D57-1 — Formal proof of Lemma C1- V_4 , resolved by D60). **Fully resolved by D60 [5].** Document D60 proves, from axioms C1 and C2 alone, that the physical indistinguishability of $s^{(1)}$ and $s^{(2)}$ in the pulsation 2-cycle forces V_4 -invariance of all physical observables, giving effective group $G_{\text{eff}} = S_4/V_4 \cong S_3$. The proof proceeds via: (i) $s \rightarrow -s$ breaks C2 (negative result, verified exhaustively); (ii) orbit-4 $\cong V_4$ as a regular V_4 -set; (iii) the three pairings are partitions into cosets of H_1, H_2, H_3 ; (iv) C1-admissibility $\Leftrightarrow V_4$ -invariance on all of $\text{Coh}(K_4)$. Two verification scripts (PDL_C1V4_script1.py, PDL_C1V4_script2.py) deposited with D60. **Entry point:** D60 [5].

Open Problem 16.19 (OP-D57-2 — Dic_3 as the weak gauge group). Identify $\text{Dic}_3 \subset \text{SU}(2)$ not merely as a finite subgroup but as the structural generator of the weak gauge group within PDL, via a universality argument from C1–C4. **Priority:** medium. **Entry point:** D33, D46, D57.

Resolved Problem 3 (OP-SU3 / OP-D57-3 — Derivation of $\text{SU}(3)$ from C1–C4, resolved (algebraic part) by D58). **Resolved (algebraic part) by D58 [75].** Document D58 derives $\text{SU}(3)$ as an unconditional algebraic theorem of C1–C4 via five lemmas and the Cartan–Killing–Weyl classification of compact simply

connected simple Lie groups. Combined with D46 and D57, the algebraic structure $SU(3) \times SU(2) \times U(1)$ of the Standard Model gauge group is an unconditional theorem of C1–C4. The physical identification of $SU(3)$ with the colour gauge group $SU(3)_c$ acting on the proton triplet (u, u, d) in the fundamental representation **3** remains open as OP-D58-1 below.

Resolved Problem 4 (OP-D58-1 — Physical identification of $SU(3)_c$, resolved by D59). *Resolved by D59 [76]. The fundamental representation **3** of $SU(3)$ is constructed on the trace-zero plane $W = \{a + b + c = 0\} \subset \mathbb{C}^3$, with axes labelled by $V_4 \setminus \{e\}$. The isospin asymmetry $\Delta n = 4 > 0$ (D47, unconditional theorem of C4) selects the orientation **3** over $\bar{\mathbf{3}}$, identifying the matter sector from the axioms without additional input. The physical colour gauge group $SU(3)_c$ acting on W in **3** is an unconditional theorem of C1–C4. **Entry point:** D59.*

Open Problem 16.20 (OP-D58-2 — The three triplets of K_4). Establish formally the relationship between the three triplets of K_4 : (T1) three independent cycles in $H_1(K_4; \mathbb{Z})$ (D51); (T2) three leakage cycles with prime exponents (23, 67, 997) (D51, D52); (T3) three V_4 -orbits on the six edges of K_4 (D58, Lemma L2). Note that (T1) and (T3) carry inequivalent representations of S_4 : V_4 acts faithfully on (T1) but trivially on (T3), so a naive bijection is excluded (D58, negative result). **Priority:** medium. **Entry point:** D51, D52, D58.

Resolved Problem 5 (OP-D58-3 — Fundamental representation **3** from C1–C4, resolved by D59). *Resolved by D59 [76]. The derivation of **3** from C1–C4 is the content of the main theorem of D59. The carrier space W , the A_2 root system in W , and the orientation forced by $\Delta n = 4 > 0$ are all unconditional theorems of C1–C4 + Cartan–Killing–Weyl. **Entry point:** D59.*

Open Problem 16.21 (OP-D59-1 — Quark mass spectrum from C1–C4). The carrier space W and the orientation **3** for $SU(3)_c$ are established by D59. The individual quark masses $m_u, m_d, m_s, \dots$ are not derived from C1–C4. The sole irreducible external parameter is $\Delta m_{\text{iso}} = m_d - m_u = 2.532 \text{ MeV}$ (D31); the derivation of individual quark masses requires additional structural input not yet identified in the axioms. **Priority:** medium. **Entry point:** D31, D47, D59. **Note v29:** Partially addressed by D63 [78]. Conjecture H_{mass} derives $m_d/m_u = 2401/1104$ from C1–C4 with no free parameter beyond Δm_{iso} ; Conjecture H_{sea} addresses m_s, m_c, m_b, m_t within 2.5% of PDG. Formal proofs from C1–C4 remain open (OP-D63-1, OP-D63-2).

Resolved Problem 6 (OP-D59-2 — Minimal covariant derivative as a C4 survival theorem, resolved by D61). *Fully resolved by D61 [4]. The minimal covariant derivative $D_\mu = \partial_\mu - ig A_\mu$ is derived from C4 via the following causal chain. (α) C4 applied to discrete transport rules on K_4 selects exactly $\text{Coh}(K_4) \cong \mathbb{Z}_2^3$ as the group of admissible transports (2048 exhaustive checks). (β) V_4 acts faithfully on $\text{Coh}(K_4)$ by group automorphisms, with fixed-point subgroup $F \cong V_4$ labelled by the three perfect matchings of K_4 (288 checks); the semidirect product $\mathbb{Z}_2^3 \rtimes V_4$ of order 32 is well-defined. (γ) The equivariant connected completion of $\mathbb{Z}_2^3$ under the V_4 -constraint is consistent with $G_{\text{gauge}} = SU(3) \times SU(2) \times U(1)$, established independently as a theorem by D46–D60. (δ) The Utiyama (1956) uniqueness theorem identifies D_μ as the unique first-order covariant operator; local gauge invariance is a C4 survival condition, not a postulate. Two verification scripts deposited with D61. **Entry point:** D61 [4].*

Open Problem 16.22 (OP-D61-1 — Gauge boson masses from C1–C4). *Partially resolved by D62 [77]. The ratio $M_Z/M_W = 1/\cos(19\pi/119)$ is now an unconditional theorem of C1–C4 (D62, Theorem 5.1). The electroweak vev v and the individual masses M_Z, M_H are derived as motivated conjectures (D62, Conjecture T1 and Conjectures 4 and 6.1). Remaining open: The covariant derivative D_μ encodes the form of the gauge coupling but not the masses of the gauge bosons $W^\pm$ and Z^0 . The derivation of $M_W/M_Z = \cos(19\pi/119)$ from C1–C4 (OP10-c) and the identification of the mass-generation mechanism within PDL remain open. **Priority:** high. **Entry point:** D55, D57, D61.*

Open Problem 16.23 (OP-D61-2 — Three fermion generations from C1–C4). The gauge structure (group, representations, dynamics) is complete. The derivation of three dynamical generation replicas from C1–C4 (OP-OFN-1) is not yet achieved. The evidence from $\beta_1(K_4) = 3$, $A_4/V_4 \cong \mathbb{Z}_3$, and the triplet

structure of the orbit $\mathcal{O}_5$ in D61 is suggestive but not a proof. **Priority:** high. **Entry point:** N01, D57, D59, D61.

Open Problem 16.24 (OP-D61-3 — Autonomous derivation of G_{gauge} from the transport structure). Arrow γ of the D61 causal chain is currently a consistency result relying on D46–D60. An autonomous derivation would establish G_{gauge} directly from the transport group $\mathbb{Z}_2^3 \rtimes V_4$ without invoking the prior algebraic route, by constructing the explicit representation of $\mathbb{Z}_2^3 \rtimes V_4$ on $\mathcal{H}_{\text{cyc}} \cong \mathbb{C}^2$ and showing that the unique connected Lie group extending this representation is G_{gauge} . **Priority:** medium. **Entry point:** D33, D46, D57–D60, D61.

Open Problem 16.25 (OP-D62-1 — Formal derivation of the correction k_1/N_{tot}^2 on v). Conjecture T1 of D62 [77] gives $v = N_{\text{tot}}^2/(R_e k_1) \cdot m_p$ at 676 ppm. A correction factor $k_1/N_{\text{tot}}^2 = 9/14161$ reduces the deviation to 41 ppm; its derivation from C1–C4 is missing. **Priority:** high. **Entry point:** D55, D57, D61, D62.

Open Problem 16.26 (OP-D62-2 — Derivation of the Higgs self-coupling λ_H from C1–C4). Derive $\lambda_H = (M_H/v)^2/2$ from C1–C4. Prerequisite: the top quark Yukawa coupling $y_t = \sqrt{2} m_t/v$; D63 H_{sea} provides $m_t \approx 176169$ MeV, giving $y_t \approx 0.947$ (5% of SM value). **Priority:** high. **Entry point:** D62, D63.

Open Problem 16.27 (OP-D62-3 — Uniqueness of the M_Z self-consistency fixed point). Prove analytically that the self-consistency equation $M_Z = g_Z[\Delta\alpha(M_Z)] \cdot v/2$ has a unique fixed point and that Newton iteration converges from any physically reasonable initial value. **Priority:** medium. **Entry point:** D62.

Open Problem 16.28 (OP-D62-4 — Hadronic vacuum polarisation corrections to $\Delta\alpha(M_Z)$). The 3.67σ tension on M_Z (D62) is traced to the absence of pion and kaon corrections to $\Delta\alpha_{\text{had}}$. D63 H_{sea} provides $m_s \approx 93.1$ MeV, enabling a Gell-Mann–Oakes–Renner estimate of m_π , m_K , and subsequently $\Delta\alpha_{\text{had}}$. Resolution is expected to reduce the tension below 2σ . **Priority:** high. **Entry point:** D62, D63.

Open Problem 16.29 (OP-D62-5 — Combinatorial origin of the factor 5^2 in the Higgs cross-section estimate). D62 invokes a factor $5^2 = (\sqrt{5})^4$ in the Higgs production cross-section estimate. Its derivation from a matching polynomial on K_4 involving the golden ratio $\varphi = (1 + \sqrt{5})/2$ is conjectured but not proved. **Priority:** medium. **Entry point:** D62.

Open Problem 16.30 (OP-D63-1 — Formal independence of C2, C3, C4 in $\mathcal{Q}(K_n)$). D63 argues that exactly $k = 3$ entity levels are required in the coherence-pulsation quadruplet $\mathcal{Q}(K_n)$ via a minimality argument. A complete proof requires showing that the constraints of C2, C3, and C4 on $\mathcal{Q}(K_n)$ are mutually independent within K_n . Resolution would elevate Conjecture H_{mass} to an unconditional theorem of C1–C4. **Priority:** high. **Entry point:** D47, D59, D63.

Open Problem 16.31 (OP-D63-2 — Formal derivation of the residual-fluidity weighting $w(n) = R_{\text{sea}}/(R_{\text{sea}} + r(n))$). The weighting $w(n) = R_{\text{sea}}/(R_{\text{sea}} + r(n))$ was selected by exhaustive numerical scan over 5 candidate functions (D63) and admits a natural structural reading (residual fluidity of the sea), but its derivation from C1–C4 applied to pseudo-stationary regimes with finite lifetime is missing. Resolution would place Conjecture H_{sea} on the same axiomatic footing as H_{mass}. **Priority:** medium. **Entry point:** D61, D63.

Open Problem 16.32 (OP-D63-3 — Structural origin of the effective sizes (52, 92, 120, 332) and of the identity $n_u - 1 = p_{k_1}$). The effective sizes of the sea quark pseudo-stationary regimes determined by retro-analysis in D63 satisfy $n_c = p_{k_1} \times 4 = 23 \times 4$ and $n_b = 5 n_u$. The identity $n_u - 1 = p_{k_1} = 23$ (D47 + D51) connects the valence quark mass hierarchy to the cosmological leakage sector by two independent unconditional theorems of C1–C4 with no known common causal origin. A structural derivation of the admissible effective sizes from the axioms would close OP-D59-1 in full generality. **Priority:** medium. **Entry point:** D47, D51, D63.

Open Problem 16.33 (OP-E2-PDL — Identification of the $E2$ operator in the PDL formalism). Identify, within the PDL signed-graph formalism, the combinatorial object corresponding to the nuclear electric quadrupole ($E2$) transition operator. Specifically: show from C1–C4 that the $E2$ matrix element is

proportional to the number N_{mix} of $(A) \wedge (B)$ -stable mixed triangles. Combined with Theorem 10.7 and D32 Prop. 3 ($N_{\text{mix}} \propto R_{\text{surf}}$), this would elevate Conjecture H_B to an unconditional theorem. **Priority:** high. **Entry point:** D29, D32, D41 [60], D56 [3].

Open Problem 16.34 (OP-D64-1 — Macroscopic microstate counting: the $K_4 \leftrightarrow K_4$ coupling rule). Derive, from C1–C4 alone, the many-nucleon entropy law of a macroscopically structured PDL assembly, equivalent in difficulty to Open Problem OP1 of D38 but for a genuinely composite closure rather than an isolated one. Proposition 7.3 (D64) shows that the naive combinatorial pairing of engaged nucleons fails the Bekenstein–Hawking target by 36.7 orders of magnitude, mass-independently; the surface-restricted refinement of D64 narrows this for one solar mass but fails to generalise to M87*. The central missing structural ingredient, identified explicitly in the corpus but not yet constructed, is a $K_4 \leftrightarrow K_4$ coupling rule: an axiomatic prescription for how two or more engaged K_4 -closures (or their proton/neutron composites) jointly contribute degrees of freedom when coupled to one another at macroscopic multiplicity, as opposed to the single-partner $(A) \wedge (B)$ coupling of D29 on which every current PDL result depends. Resolving this would also open the quark–gluon plasma regime near black hole formation and is identified as the single highest-priority entry point for the continuation of the black hole physics programme. **Priority:** highest. **Entry point:** D29, D37, D38, D64, D65.

Open Problem 16.35 (OP-D64-2 — A fully relational unit system for c and μ^*). Construct a unit system in which the speed of light c and the proton–electron mass ratio μ^* are derived from C1–C4 rather than imported from measurement, as a necessary precondition for an autonomous PDL derivation of $\lambda_{\text{PDL}} = 4\ell_P^2$ (the principal open problem of D37). As established in the corpus (D64, and recalled throughout this document as a key learning), c and $\hbar$ are translation factors converting PDL’s discrete combinatorial register into continuous physical measurement, and treating their derivation as a missing theorem is a category error for the constants themselves; what remains genuinely open is not c or $\hbar$ as such, but the 47 ppm residual in the mass-ratio formula μ (Open Problem 16.8, OP7), to which this problem is closely related. **Priority:** medium. **Entry point:** D28–D30, D37, D64.

Open Problem 16.36 (OP-D65-1 — The shared S^2). Is the topological type S^2 , appearing independently in the origin of the exponent 18 (D23 v2, proton scale), in the forced two-defect theorem (Theorem 7.4, horizon scale), and in the near-horizon $AdS_2 \times S^2$ geometry of extremal Kerr [81], more than a shared topological vocabulary? Specifically: does a single combinatorial construction on K_4 exist from which all three follow as corollaries, rather than as three independently derived consequences of the same manifold type? **Priority:** high. **Entry point:** D23 v2, D65.

Open Problem 16.37 (OP-D65-2 — Combinatorial-to-physical time near extremality). Does the tension identified in Theorem 7.5 (finite combinatorial time to reach $\chi_f = 1$, in apparent tension with the third law of black hole thermodynamics) resolve because the mapping from combinatorial time to physical time itself diverges near extremality, or does it reflect a genuine, falsifiable departure of this specific PDL mechanism from the third law? A resolution requires either an independent, non-circular derivation of this map beyond the single anchor GM/c^3 used in D65, or an explicit demonstration that no such divergent map exists, which would itself constitute a distinctive prediction. **Priority:** high. **Entry point:** D65.

Open Problem 16.38 (OP-D65-3 — A PDL theory of the inhomogeneous, symmetry-broken regime). Construct the coherence stress-energy tensor C_{coh} , the equation of state, and the local curvature bound of D48v3 and D54 for a macroscopically structured closure with a preferred axis (finite inter-cell coherence-variation scale L), rather than in the homogeneous limit $L \rightarrow \infty$ on which every current theorem in that framework depends. This is identified in D65 as the common cause of three of the seven negative results of Remark 7.7, and is proposed as the single most consequential open problem for future PDL work on black holes, superseding the separate pursuit of any one of its symptoms. **Priority:** highest. **Entry point:** D48v3, D54, D65.

Open Problem 16.39 (OP16 — ZBM3–PDL correspondence). Establish a formal correspondence between the Zero-Body Matrix Elements (ZBM3) of the DNO-SM shell-model framework and the PDL

closure-multiplicity counting. D41 conjectures this correspondence on the basis of the 2% agreement in the wave-function component ratio. **Priority:** medium. **Entry point:** D41.

Open Problem 16.40 (OP17 — Falsifiable predictions for $^{88}\text{Ru}/^{90}\text{Ru}$ and $^{92}\text{Pd}/^{94}\text{Pd}$). Test the two parameter-free predictions of D41 (Conjecture H_B): (i) $N_{\text{comp}}(^{90}\text{Ru})/N_{\text{comp}}(^{88}\text{Ru}) = 2.000 \pm 0.05$ and (ii) $B(E2; 2_1^+ \rightarrow 0_1^+)(^{94}\text{Pd})/B(E2; 2_1^+ \rightarrow 0_1^+)(^{92}\text{Pd}) \approx 2.02$ at FRIB or RIKEN. **Priority:** experimental. **Entry point:** D41.

Resolved Problem 7 (OP-pressure — Equation of state of the PDL coherence fluid, resolved by D54). *The equation of state $w(N) = -\rho_\Lambda/\rho_{\text{coh}}(N)$ of the PDL coherence fluid is an unconditional theorem of C1–C4 (D54 [1]). In the homogeneous ground state: $w_{\text{mass}} = 0$ (D48v3 Thm 4.1); $w_{\text{spin}} = 0$ (D48v3 Thm 5.3); $w_{\text{orb}} = 0$ (D49 Thm 1); $w_{\text{leak}} = -1$ (D48v3 Prop. 8.1). The PDL dark-energy fraction $\Omega_\Lambda^{\text{PDL}} = 0.6838$ agrees with Planck 2020 to 0.17% with no cosmological parameter as input. The coherence fluid interpolates between cold dark matter ($|w| \sim 10^{-44}$) and dark energy ($w \approx -0.62$) as a single combinatorial structure.*

Open Problem 16.41 (OP-DM — Identification of the coherence fluid with observed dark matter). D54 proves that the PDL coherence fluid has $w \approx 0$ at nuclear and sub-nuclear densities, identical in equation of state to cold dark matter. Whether the fluid can be identified with the observed dark matter component of the universe — i.e. whether $\rho_{\text{coh}}(N)$ integrated over cosmological scales accounts for the observed matter density $\Omega_m \rho_{\text{crit}}$ — requires a dedicated cosmological analysis. **Priority:** medium. **Entry point:** D24, D27, D35, D54.

Open Problem 16.42 (OP-inhomogeneous — Equation of state beyond the ground state). Theorem 1 of D54 applies to the homogeneous ground state ($\nabla\phi = 0$ over V_C , $j^i = 0$). In the inhomogeneous regime, or in the presence of an external electromagnetic field, P_{orb} and P_{spin} are non-zero. Deriving the full equation of state in that regime from C1–C4 requires specifying the inter-cell distribution of the quantum state ψ . **Priority:** medium. **Entry point:** D48v3, D49, D54.

Resolved Problem 8 (OP-spin — Spin component of C_{coh} , resolved in D48v3). *The spin component $C_{\text{spin}\mu\nu}$ of the PDL coherence tensor is $C_{\mu\nu}^{(\text{spin})} = 0$ in the homogeneous ground state. Proof: explicit computation over all 8 coherent configurations of K_4 gives $\langle S^i \rangle = 0$ (Lemma 5.1 of D48v3) and $\langle S^i S^j \rangle = \frac{1}{4} \delta^{ij}$ (Lemma 5.2 of D48v3). The traceless part of the Weyssenhoff–Raabe spin tensor vanishes exactly, and the isotropic gradient term vanishes by the London gauge theorem of D49 (C4 forces uniform phase over V_C). This is an unconditional theorem of C1–C4. Resolved in D48v3 [67] (DOI 10.5281/zenodo.20151380).*

Open Problem 16.43 (OP-London — Resolved: London equation from PDL axioms). **Fully resolved by D49.** The London equation $j_s^i = -(n_{\text{coh}} e^2 / m_p c) A^i$ is derived from C1–C4 via: Lemma 1 ($\nu = \frac{1}{4} \|\psi_1 - T^k \psi_1\|^2$, 768/768, exact integer arithmetic); Theorem 1 (C4 forces $\psi_j = \psi_1$ on V_C via C3 connectivity, hence $\phi = \text{const}$, London gauge); Corollary (minimal substitution from D32+D46 gives the London equation with $n_{\text{coh}} = \sigma(N) R_{\text{surf}} / (V_C R_{\text{tot}}) \in \mathbb{Q}(\sqrt{5})$).

17 Logical Dependency Map

The following critical path summarises the logical flow of the programme from axioms to falsifiable predictions. Each arrow $A \Rightarrow B$ means: the proof of B requires A as a direct input.

18 Logical Dependency Map

The following two figures summarise the complete logical flow of the PDL programme. Figure 1 covers the foundational causal chain from axioms to physical constants and dynamics. Figure 2 covers the gauge sector in full detail, from K_4 through the Standard Model gauge group to gauge dynamics. Each arrow $A \Rightarrow B$ means: the proof of B requires A as a direct input.

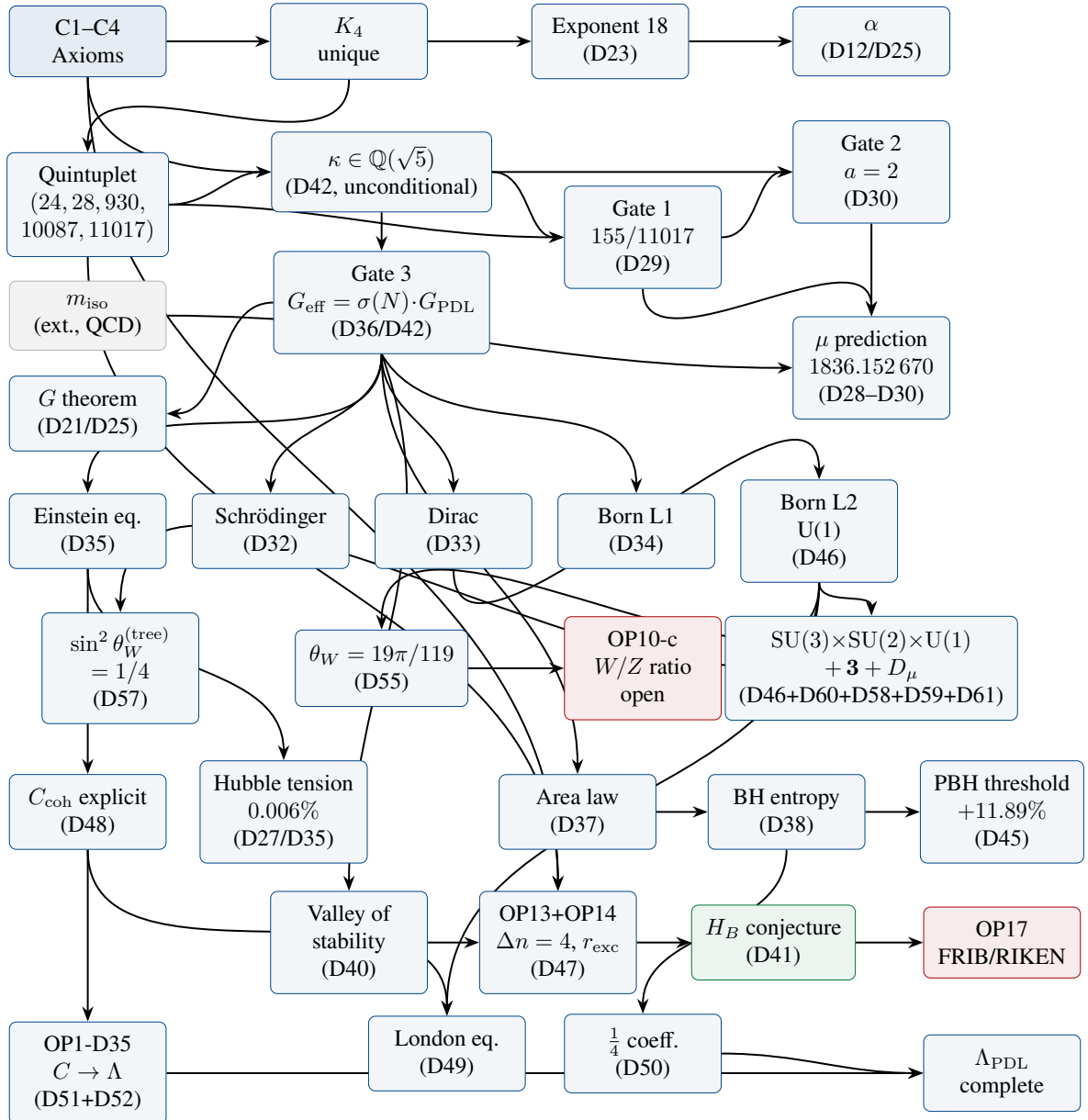
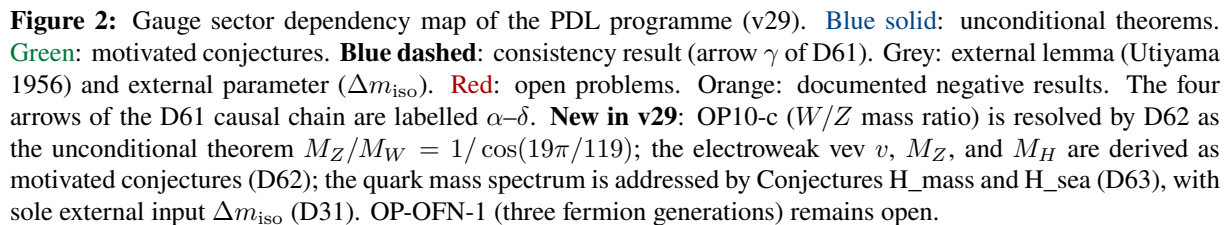


Figure 1: Critical-path dependency map of the PDL programme — foundational chain (v28). **Blue:** unconditional theorems. **Green:** conjectures. **Red:** open problems. **Grey:** unique external input (m_{iso}). The quantum dynamics layer (Schrödinger, Dirac, Born, London), nuclear stability, black hole thermodynamics, and cosmological constant are complete theorems of C1–C4. The gauge sector node (bottom right) is a summary; the full derivation chain $K_4 \rightarrow D_\mu$ is detailed in Figure 2.



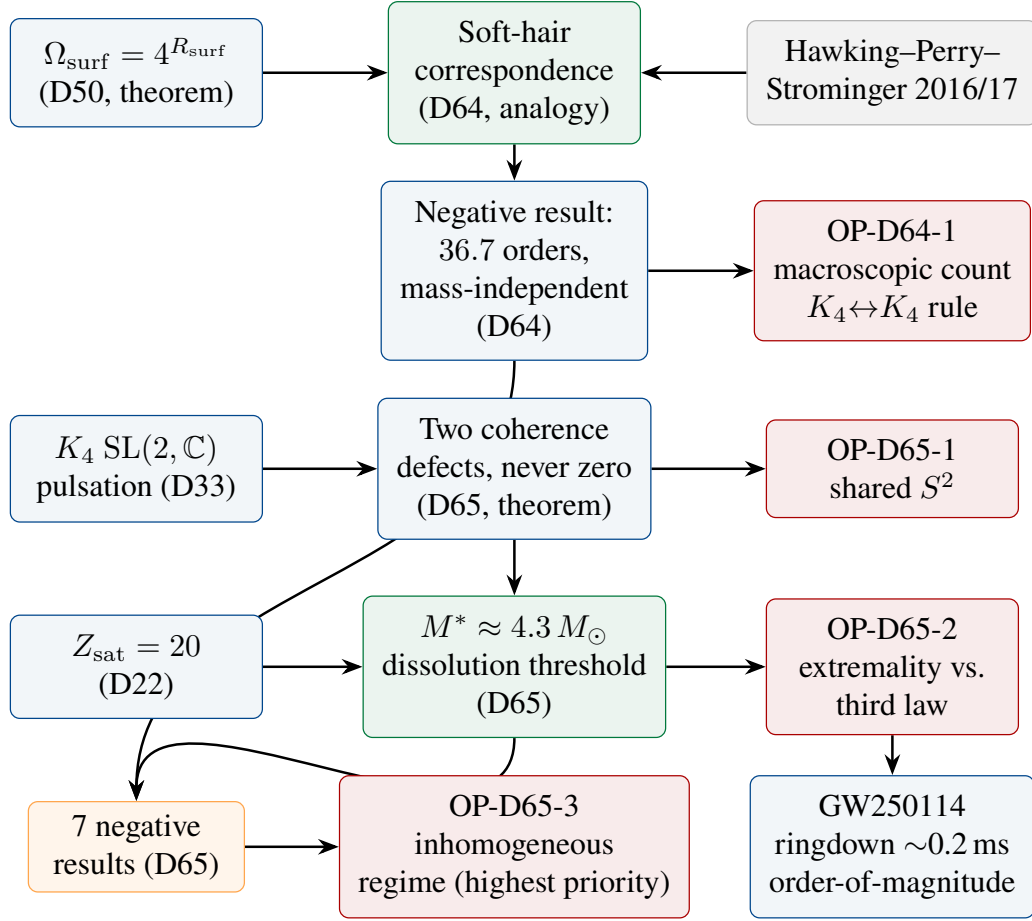


Figure 3: Black hole physics II dependency chain (new in v30). **Blue:** unconditional theorems. **Green:** conjectures. **Orange:** documented negative results. **Red:** open problems. **Grey:** external literature (soft hair). The programme moves from the D50 surface-degeneracy theorem, through the D64 soft-hair analogy and its honest negative result, to the D65 behavioural theorems (two coherence defects, curvature-migration exponent, dissolution threshold) and their associated open problems. OP-D64-1 and OP-D65-3 are jointly the highest-priority entry points.

19 Falsifiable Predictions

Table 3 Complete list of falsifiable, parameter-free PDL predictions (v30). All predictions follow from the quintuplet and Δm_{iso} alone, with no additional free parameters. Predictions P1–P6 are derived from the analytical chain; P7–P8 follow from Conjecture H_B (D41); P9 is derived from D45; P10 from D55; P11–P17 from D62 and D63; P18–P19 from D45 v2 and D65 (new in v30).

ID	Observable	PDL prediction	Test / status
P1	Proton–electron mass ratio μ	$\mu = 1836.152\,670$ (parameter-free)	Testable at 0.001 ppm; current CODATA: 1836.152 673 43(11); residual 1.8×10^{-9} .
P2	Fine-structure constant α	$\alpha^{-1} = 137.035\,999$ from quintuplet alone	Testable at 10^{-9} ; current agreement $\mathcal{O}(10^{-4})$; OP3/OP7 for residual.
P3	Hubble constant ratio	$H_{0,\text{local}}/H_{0,\text{CMB}} = \sigma(N_{\text{local}})/\sigma(40)$	PDL prediction 1.085935 vs. observed 1.0859 ± 0.007 ; 0.006% error; zero free parameters.
P4	CMB-epoch BH entropy suppression	Factor $\sigma(40) \approx 0.848$ relative to local universe	No direct observation yet; accessible via future CMB- S_4 constraints.
P5	PBH survival threshold	$M_{\text{PDL}}^* = \sigma(40)^{-2/3} M_{\text{GR}}^* \approx 5.706 \times 10^{14} \text{ g } (+11.89\%)$	Testable via Fermi-LAT IGRB analysis with BLACKHAWK/ISATIS [64]; excess Hawking flux at $\sim 93\text{--}104 \text{ MeV}$.
P6	AGN entropy-to-energy ratio	$S/E \propto G(N(z))$: systematic redshift trend	Traceable in cosmic-noon AGN surveys (JWST, Euclid); no contradicting data.
P7	$B(E2)$ ratio in $^{90}\text{Ru}/^{88}\text{Ru}$	$R \approx 2.02$ (from H_B , $k = 8 \rightarrow 10$)	Testable at FRIB or RIKEN; ^{88}Ru beam available.
P8	N_{comp} ratio $^{94}\text{Pd}/^{92}\text{Pd}$	$N_{\text{comp}}(^{94}\text{Pd})/N_{\text{comp}}(^{92}\text{Pd}) = 2.000 \pm 0.05$	Testable from DNO-SM calculations with ^{94}Pd spectroscopy at RIKEN.
P9	IGRB spectral cut-off position	$E_c^{\text{PDL}} \approx 93 \text{ MeV}$ vs. $E_c^{\text{GR}} \approx 104 \text{ MeV}$ (shift -10.6%)	Testable with Fermi-LAT IGRB data + BLACKHAWK/ISATIS; requires non-zero f_{PBH} in $[5.10, 5.71] \times 10^{14} \text{ g}$.
P10	Isospin mass splitting Δm_{iso}	$\Delta m_{\text{iso}} = 2.446 \text{ MeV}$ (from D55 C1+C3 formula, no free parameter)	FLAG 2024: $2.52 \pm 0.08 \text{ MeV}$; compatible at 0.92σ . Distinguishable from D30 value (2.532 MeV) at $\pm 0.04 \text{ MeV}$ precision (next-generation lattice QCD, 3–5 year horizon).

ID	Observable	PDL prediction	Test / status
P11	Electroweak vev v	$v = 14161/54 \cdot m_p = 246.20 \text{ GeV}$ (Conjecture T1, D62)	PDG: 246.22 GeV; deviation 676 ppm. Entry point: OP-D62-1 for the correction to 41 ppm.
P12	Z^0 boson mass	$M_Z = 91.168 \text{ GeV}$ via analytical chain $\Delta\alpha(M_Z)$ (D62)	PDG: 91.1876 GeV; tension 3.67σ ; traced to absence of hadronic corrections (OP-D62-4).
P13	Higgs boson mass	$M_H/m_p = 133.611$, i.e. $M_H = 125.33 \text{ GeV}$ (D62)	PDG: $125.20 \pm 0.11 \text{ GeV}$; tension 1.49σ .
P14	Valence quark mass ratio	$m_d/m_u = 2401/1104 \approx 2.1748$ (H_mass, D63)	PDG: 2.162 ± 0.063 ; deviation 0.59%; no free parameter beyond Δm_{iso} .
P15	Up and down quark masses	$m_u = 2.155 \text{ MeV}$, $m_d = 4.687 \text{ MeV}$ (H_mass + D31, D63)	PDG: $2.16 \pm 0.07 \text{ MeV}$, $4.67 \pm 0.15 \text{ MeV}$; deviations 0.22% and 0.37%.
P16	Strange quark mass	$m_s = 93.08 \text{ MeV}$ (H_sea, $n_s = 52$, D63)	PDG: $93.5 \pm 0.8 \text{ MeV}$; deviation 0.45%.
P17	Charm, bottom, top quark masses	$m_c = 1301 \text{ MeV}$ ($n_c = 92$), $m_b = 4081 \text{ MeV}$ ($n_b = 120$), $m_t = 176169 \text{ MeV}$ ($n_t = 332$) (H_sea, D63)	PDG: 1270 ± 30 , 4180 ± 30 , $172760 \pm 720 \text{ MeV}$; deviations 2.46%, 2.37%, 1.97%; top non-hadronisation structural ($r(332) \gg R_{\text{sea}}$).
P18	PBH photon spectral peak	$E_{\text{peak}} = 117.5 \text{ MeV}$ (PDL) vs. 130.1 MeV (GR), -9.65% shift (D45 v2, GAMMAPBHPlotTER/BLACKHAWK)	Independent spectral signature alongside the $+11.89\%$ mass-threshold shift (P5); testable with existing Fermi-LAT IGRB data.
P19	Compact-object mass gap	Nucleon-dissolution threshold $M^* \approx 4.3 M_\odot$ (Conjecture, D65), from $Z_{\text{sat}} = 20$ (D22) alone	Falls inside the observed lower mass gap; not tuned to do so. Sensitivity scan: $M^* \in [2.5, 5.0] M_\odot$ for nuclear force range $r_c \in [1.39, 2.21] \text{ fm}$, overlapping the pion Compton wavelength.

20 Continuation Guide

20.1 Entry points

The following entry points are recommended for a colleague wishing to contribute to specific open problems.

- **OP1-D35 (leak constant C):** The most significant remaining foundational problem. Requires familiarity with D48 (coherence tensor), D17 (exponent 18 derivation), and D30 (η_L structural factor). The question is whether the prefactor C in $C^{(\text{leak})}$ can be expressed as a closed-form function of the quintuplet, in the spirit of OP-B (resolved in D44).

- **OP2 (global uniqueness):** Requires familiarity with D16, D16b, and combinatorial optimisation over integer lattices. The Colab infrastructure of D43 can be used to extend the computational search radius.
- **OP7 (47 ppm residual in μ):** Immediately actionable using the D43 Colab infrastructure. Start with the Group I filters in D43, identify the residual at each stage, and test candidate combinatorial factors from C1–C4.
- **OP12 (BH coefficient $\frac{1}{4}$):** [Resolved by D50](#). The coefficient $\frac{1}{4}$ in $S_{\text{BH}} = k_B A / (4\ell_P^2)$ is the stable fraction $4/16$ per surface relation (D29), with independence guaranteed by H3 (D42). $\Omega_{\text{surf}} = 4^{R_{\text{surf}}}$, $S_{\text{surf}} = k_B R_{\text{surf}} \ln 4$. Black hole thermodynamics layer complete.
- **OP-London (London equation from PDL):** [Resolved by D49](#). See corpus table entry above.
- **OP1-D35 (leak constant C):** [Resolved by D51+D52](#). See corpus table entry and Section 16 above.
- **OP-D64-1 (macroscopic microstate counting; new in v30):** The single highest-priority entry point in the programme. Requires constructing a $K_4 \leftrightarrow K_4$ coupling rule generalising the single-partner $(A) \wedge (B)$ criterion of D29 to macroscopic multiplicity. Start from the negative result of D64 (Proposition 7.3) and the surface-restricted refinement, understanding precisely why the latter fails to generalise from one solar mass to M87*.
- **OP-D65-3 (inhomogeneous, symmetry-broken regime; new in v30):** Requires extending the coherence stress-energy tensor (D48v3) and equation of state (D54) beyond the homogeneous limit $L \rightarrow \infty$. Identified in D65 as the common structural cause of three independent negative results; resolving it would remove the single largest obstacle to a quantitative black hole physics programme.

20.2 Recommended next documents

Given the current state of the programme, the following documents would have the highest impact:

1. **DM v30:** The present document. Incorporates D64 (soft-hair correspondence, honest negative result) and D65 (behavioural universality between the proton and the black hole horizon, two-coherence-defect theorem, dissolution mass threshold), and records the outcome of PDL–OFN Session 67 (Section 14) and the current status of Note N02.
2. **The $K_4 \leftrightarrow K_4$ coupling rule (OP-D64-1):** The central missing structural ingredient blocking OP-D64-1, OP-D65-3, and the quark–gluon plasma regime near black hole formation. Highest priority of the entire programme at present.
3. **D-Kerr-extension:** Apply the D65 closure results (two coherence defects, curvature-migration exponent) systematically to the GW250114 reference target ($M_f = 62.7 M_\odot$, $\chi_f = 0.68$), extending beyond the order-of-magnitude ringdown estimate of Section 7.3.
4. **N02 finalisation:** Complete *From $\mathbb{Z}_3$ to Three Generations* with Oleg Evdokimov; the PDL-side draft (Section 14) is complete and awaits the OFN-side contribution.
5. **OP-D63-1 — Formal independence in $\mathcal{Q}(K_n)$:** Resolution would elevate H_mass to an unconditional theorem. **Entry point:** D47, D59, D63.
6. **OP-D62-4 — Hadronic corrections to M_Z :** D63 provides $m_s \approx 93.1$ MeV as the necessary input. **Entry point:** D62, D63.
7. **OP-D61-2 / OP-OFN-1 — Three fermion generations:** Formalise the link between $\beta_1(K_4) = 3$, $A_4/V_4 \cong \mathbb{Z}_3$, orbit $\mathcal{O}_5$, and three dynamical generation replicas. **Entry point:** N01, N02, D57, D59, D61, Section 14.

8. **D-electroweak-WZ**: Derivation of the W/Z mass ratio $M_W/M_Z = \cos(19\pi/119)$ as a parameter-free prediction. **Entry point**: D55, D57, D61.
9. **Experimental confrontations**: (i) FLAG/lattice QCD for Δm_{iso} at ± 0.04 MeV (D55, P10). (ii) D45/D45 v2 predictions to BLACKHAWK/ISATIS community, Fermi-LAT IGRB 50–200 MeV window (mass threshold and spectral-peak shift). (iii) Arbey/Auffinger (BLACKHAWK/ISATIS) and Recchia/Lenzi (nuclear spectroscopy, D47 predictions at FRIB/RIKEN) outreach follow-up.
10. **D-nuclear-Z>82**: Extension of the valley-of-stability theorem to $Z > 82$ (OP15).
11. **DL03**: Numerical enclosure of n_{life}^* from the gain function R_{active}^1 .

21 Glossary

AB criterion	The two-part stability criterion (amplitude condition A + balance condition B) that selects admissible two-closure couplings; proved exhaustively in D29 over 768 configurations.
Active surface R_{surf}	The set of valence coherence relations of the proton, $ R_{\text{surf}} = \sigma = 930$; the locus of all physical information exchange.
Admissible closure	A finite signed graph satisfying all four axioms C1–C4; the minimal example is K_4 .
Born Level 1 / Level 2	Level 1 (D34, Theorem): modulus rule $P = \psi ^2$ proved via Gleason uniqueness. Level 2 (D46, Theorem, OP4 resolved): $U(1)$ phase freedom derived from K_4 sign equivalence and Hopf fibration structure.
C_{coh}	Coherence stress-energy tensor appearing in the PDL Einstein equation; explicit form derived in D48v3, D51, D52: $C^{(\text{mass})}$, $C^{(\text{orb})}$, $C^{(\text{spin})}$, and $C^{(\text{leak})}$ all unconditional theorems. $C^{(\text{spin})} = 0$ in the ground state (OP-spin resolved, D48v3). All components in uniform units J m^{-3} . OP2-D35, OP1-D35, and OP-spin fully resolved.
Behaviour-versus-magnitude principle	The methodological claim, new in v30, that at the black hole scale PDL currently supports claims about topologically forced configurations, required functional forms, and threshold conditions (<i>behaviour</i>), but not yet absolute entropy values or coupling strengths (<i>magnitude</i>); proposed as a structural property of the theory's present domain of validity, not a temporary gap (D65).
Coherence defect	A point of the internal state space $\mathcal{H}_{\text{cyc}}$ where the coherent pulsation structure of K_4 cannot extend continuously over a closed surface; exactly two are forced on any closed spherical PDL surface, never zero, by a combinatorial analogue of the Poincaré–Hopf theorem (D65, Theorem 7.4).
Curvature-migration exponent	The exponent governing how the polar curvature deficit of a PDL closure migrates towards the equator as the spin parameter χ approaches extremality; equal to exactly 1 in arbitrary-precision arithmetic (D65), an open tension with the third law of black hole thermodynamics (OP-D65-2).

Dissolution threshold M^*

The critical mass $M^* \approx 4.3 M_\odot$ at which the density required for nucleon dissolution (from $Z_{\text{sat}} = 20$, D22, and the nuclear force range) coincides with the Schwarzschild condition; falls inside the observed compact-object mass gap without tuning (D65, Conjecture 7.6).

 $K_4 \leftrightarrow K_4$ coupling rule

The identified but not yet constructed axiomatic prescription for how two or more engaged K_4 -closures jointly contribute degrees of freedom at macroscopic multiplicity, generalising the single-partner $(A) \wedge (B)$ criterion of D29; the single highest-priority missing structural ingredient for OP-D64-1 (macroscopic microstate counting).

Soft-hair correspondence

The structural analogy, established in D64, between the PDL surface-degeneracy theorem $\Omega_{\text{surf}} = 4^{R_{\text{surf}}}$ (D50) and the Hawking–Perry–Strominger soft-hair programme; explicitly not a claim of mathematical identity.

Coherence-pulsation cost

$|\mathcal{Q}(K_n)| = r(n) \cdot n^3 = n^4(n-1)/2$: the cardinal of the coherence-pulsation quadruplet set of a valence core K_n ; governs the valence quark mass ratio via Conjecture H_{mass} (D63).

Conjecture H_{mass} $m_d/m_u = 2401/1104$: the ratio of coherence-pulsation costs of K_{n_d} and K_{n_u} ; identified by exhaustive scan of degree- ≤ 5 monomials; no additional free parameter beyond Δm_{iso} (D63).

Conjecture H_{sea} $m_{\text{sea}}(n) = [n^4(n-1)/n_u^4(n_u-1)] \cdot [R_{\text{sea}}/(R_{\text{sea}} + r(n))] \cdot m_u$: effective mass of a pseudo-stationary sea quark regime; weighting $w(n)$ selected by exhaustive scan over 5 candidates (D63).

Hadronisation boundary

$n_{\text{max}} = 140$: maximum effective size (multiple of 4) such that $r(n) \leq R_{\text{sea}}$; corresponding maximum effective mass ≈ 7677 MeV; the top quark ($n_t = 332$, $r(332) \gg R_{\text{sea}}$) lies above this boundary and cannot form hadrons (D63, structural corollary).

Residual fluidity $w(n) = R_{\text{sea}}/(R_{\text{sea}} + r(n))$: fraction of the extended relational budget $R_{\text{sea}} + r(n)$ that remains available as fluid sea when a pseudo-stationary structure of size n is present; governs Conjecture H_{sea} (D63).

Structural identity $n_u - 1 = p_{k_1} = 23$: exact integer identity between two unconditional theorems of C1–C4 — the valence up-quark size ($n_u = 24$, D47) and the first cosmological leakage prime ($p_{k_1} = p_9 = 23$, D51) — connecting the hadronic and cosmological sectors for the first time (D63).

Coherence cost Fraction of violated triangles (negative-product edges) in a signed graph; minimised by axiom C4.

DL series Documents DL01, DL02, ...; the PDL-V programme extending C1–C4 to self-replication and complexity thresholds. Structurally separate from the D series; presented in Part III of this document.

δ^* Optimal lineage variation in the PDL-V programme; $\delta^* = 1/2$, equal to the half-pulsation of K_4 ; proved invariant under the rarity law (DL01).

ε_G^B	Geometric correction factor in the PDL mass-ratio formula; closed-form expression $\varepsilon_G^B = \kappa \cdot C / (6 + \kappa) \cdot (1 + 2/R_{\text{tot}}) \in \mathbb{Q}(\sqrt{5})$ proved in D44 (OP8 resolved).
Equation of state $w(N)$	The ratio $P/(\rho_{\text{coh}}c^2)$ for the PDL coherence fluid; $w(N) = -\rho_\Lambda/\rho_{\text{coh}}(N)$ (D54, unconditional theorem). At coherence densities $ w \sim 10^{-44}$ (cold matter); at cosmological scales $\Omega_\Lambda^{\text{PDL}} = 0.6838$ (dark energy, 0.17% from Planck 2020).
$f(n)$	Rarity law in the PDL-V programme; $f(n) = 2^{n+1-\binom{n}{2}}$, doubly exponentially decreasing; exponents are triangular numbers (DL01, exact theorem).
$G_{\text{eff}}(N)$	Effective gravitational coupling at scale N ; equal to $\sigma(N) \cdot G_{\text{PDL}}$ by Gate 3 (unconditional theorem).
Gate 1, 2, 3	Three critical proof milestones. Gate 1: 155/11017 from axioms (D29). Gate 2: $a = 2$ from axioms (D30). Gate 3: $G_{\text{eff}} = \sigma(N) \cdot G_{\text{PDL}}$ (D36/D42, unconditional).
H1, H2, H3	Named hypotheses of the programme. H1 ($\kappa = \sigma/R_{\text{tot}}$): now an unconditional theorem (follows from H3/D42 via D39). H2 (bridge linearity): absorbed into D36. H3 (Indifference Lemma): proved unconditionally in D42.
Hopf fibration	The fibre bundle $S^1 \hookrightarrow S^3 \rightarrow S^2$ generating the $U(1)$ phase freedom of quantum mechanics; its appearance in D46 is a structural consequence of the K_4 pulsation sign equivalence.
Indifference Lemma (H3)	The probability measure on the relations of a PDL closure is uniform; the unique measure compatible with S_4 -symmetry of K_4 . Proved from C1–C4 in D42 via three lemmas.
K_4	The complete signed graph on four vertices and six edges; the unique minimal admissible closure (D16a theorem); identified with the electron prototype.
κ	Primary leakage parameter; $\kappa = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$; unconditional theorem of C1–C4 (D42 corollary); $\kappa \approx 0.0455$.
Mirror Lemma	The proton and neutron HO-PDL levels are isomorphic, sharing the same splitting $s = 1/12$, level order, and degeneracies (D47, Lemma 5.2, from D22-N2).
m_{iso}	Isospin mass splitting $m_d - m_u$; the unique irreducible external parameter of the programme (Gate 2, D30).
N_{CMB}	PDL-derived CMB epoch parameter; $N_{\text{CMB}} = 40$ from neutron quintuplet (D27 theorem).
$n_{\text{life}}^*, n_{\text{consciousness}}^*, n_{\text{max}}^*$	Life, consciousness, and maximum complexity thresholds in the PDL-V programme; existence and strict ordering proved (DL02); numerical values open (DL-OP1, DL-OP2).
OPn	Open Problem n ; see Section 16. OP1, OP4, OP8, OP12, OP13, OP14, OP-London, OP1-D35, and OP-pressure fully resolved.
PDL	Projective Dynamic Logo; the research programme initiated in D01.
PDL-V programme	Extension of PDL to self-replication and complexity thresholds; inaugurated by DL01 and DL02; see Part III.

Quintuplet	The integer tuple (24, 28, 930, 10087, 11017) characterising the proton; $\nu_u = 24$, $\nu_d = 28$, $\sigma = 930$, $R_{\text{sea}} = 10087$, $R_{\text{tot}} = 11017$.
$\rho_{\text{coh}}(N)$	Coherence mass density at scale N ; $\rho_{\text{coh}} = \sigma(N) \cdot R_{\text{surf}} \cdot m_p / (V_C \cdot R_{\text{tot}})$ (D48v3 theorem).
$S(\Gamma)$	Constraint-structure operator in the PDL-V programme; $S(K_4) \cong K_4$ (DL01 theorem).
$\sigma(N)(N)$	Coherence engagement fraction at scale N ; $\sigma(N) = 1 - (1 - \kappa)^N \in \mathbb{Q}(\sqrt{5})$; $\sigma(N)(40) \approx 0.8449$.
T_{pp}	Proton–proton conflict unit; $C(Z > 20) = 190 T_{pp}$ exactly (D40 theorem).
V_C	Coherence volume; $V_C = (4/3)\pi[\hbar/(m_p c)]^3$; unconditional theorem of C1–C4 via D10a (temporal chain, Chain 1). D48v3 theorem.
Valley of stability	Minimum neutron number $N_{\min}(Z)$ for stable nuclei; reproduced at 100% accuracy for $Z = 1\text{--}82$ as an unconditional theorem of C1–C4 (D40, D47 corollary).
18-filter hierarchy	The decomposition $18 = 6 + 5 + 4 + 3$ of the leakage exponent into four topological levels (D06, D17, D23).

Part I

The PDL-V Programme: Life and Consciousness

This Part is structurally separate from Parts I and II.

It presents a rigorous extension of the PDL framework to self-replication and complexity thresholds, in a domain distinct from the physical constants programme. The same axioms C1–C4 are used; the domain of application is different.

22 Introduction to the PDL-V Programme

The PDL-V programme asks whether the axioms C1–C4, which determine the proton architecture and all derived physical constants, also constrain the structural conditions for self-replication and the emergence of biological and cognitive complexity. Documents DL01 and DL02 do not make biological claims; they establish mathematical threshold theorems that any system satisfying C1–C4 must obey. The connection to biology and consciousness is conjectural and is explicitly labelled as such.

22.1 Separation from the physics programme

The DL series shares the axiomatic foundation (C1–C4 and K_4) with the D series, but operates in a domain where the “configurations” are hierarchical graph structures rather than physical coherence cycles. The results of DL01 and DL02 are theorems of C1–C4 applied to this domain, with full epistemic rigour, but their physical interpretation involves a conjecture (the PDL-V Conjecture, DL01) that is not yet derivable from first principles. This conjecture is explicitly identified and is not presented as a theorem.

22.2 Reading note

A reader whose sole interest is the physical constants programme (gravitational constant, cosmological constant, nuclear structure, etc.) may skip Part III without loss of continuity. A reader interested in the structural limits of self-replication and complexity should read DL01 first (operators L , S , R^k ; rarity law; δ^* invariant), then DL02 (threshold hierarchy theorem).

23 Self-Replication Operators and the Rarity Law (DL01)

Theorem 23.1 (K_4 fixed point of S , [8]). *Let $S(\Gamma)$ be the constraint-structure operator acting on admissible signed graphs. The complete graph K_4 is the unique fixed point of S : $S(K_4) \cong K_4$. This has been verified computationally over all 8 coherent configurations of K_4 .*

Theorem 23.2 (Rarity law and δ^* invariant, [8]). *The fraction of self-replicating configurations at level n satisfies the exact law:*

$$f(n) = 2^{n+1-\binom{n}{2}}, \quad (9)$$

with exponents equal to $-T_{n-2}$ (triangular numbers): $f(n)$ is doubly exponentially decreasing. The optimal lineage variation $\delta^ = 1/2$ is an invariant equal to the half-pulsation of K_4 ; it is independent of n . The weak self-representation relation R_{weak}^1 is forced at 100% for all $n \geq 4$, verified for $n = 4-7$.*

24 Threshold Hierarchy Theorems (DL02)

Theorem 24.1 (K_4 unique fixed point, [9]). *K_4 is the unique fixed point of S over all admissible closures. The equation $C(n, 3) = n(n-1)(n-2)/6 = 4$ has $(n-1)(n-2) = 6$ as its only positive integer solution: $n = 4$. This is an unconditional theorem of C1–C4.*

Theorem 24.2 (Strict threshold separation, [9]). *For $n \geq 5$, $|S^1(K_n)| \neq |S^2(K_n)|$: the first and second iteration of S produce distinct orbit sizes. This forces a strict separation between the life threshold and the consciousness threshold: $n_{\text{life}}^* < n_{\text{consciousness}}^*$. This is an unconditional theorem of C1–C4.*

Theorem 24.3 (Finiteness of all thresholds, [9]). *The doubly exponential rarity law $f(n) = 2^{n+1} - \binom{n}{2}$ implies that all thresholds n^* are finite: $n_{\text{max}}^* < \infty$. Combined with Theorems 24.1 and 24.2, this gives the unconditional hierarchy:*

$$4 < n_{\text{life}}^* < n_{\text{consciousness}}^* < n_{\text{max}}^* < \infty. \quad (10)$$

24.1 Open problems in the PDL-V programme

Open Problem 24.4 (DL-OP1 — Numerical value of n_{life}^*). The existence and strict ordering of n_{life}^* is established analytically (DL02). Its numerical value remains open. The gain function $R_{\text{active}}^1(n)$ must be derived in closed form and its first zero located. **Priority:** high. **Entry point:** DL01, DL02.

Open Problem 24.5 (DL-OP2 — Numerical value of $n_{\text{consciousness}}^*$). Analogous to DL-OP1 but for the second threshold. Requires DL-OP1 as a prerequisite. **Priority:** medium. **Entry point:** DL01, DL02.

Open Problem 24.6 (DL-OP3 — Connection to physical substrate). The PDL-V conjecture (DL01) proposes that the threshold n_{life}^* corresponds to a specific physical structure derivable from C1–C4. This connection has not been derived and remains a conjecture. A formal derivation would require identifying the physical interpretation of the graph complexity level n in terms of molecular or cellular organisation. **Priority:** exploratory. **Entry point:** DL01.

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